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Communication-Based Social Network Search Algorithms Are Used for Numerical Optimization and Practical Applications

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Abstract

To enhance the performance of the Social Network Search (SNS) algorithm in solving complex numerical optimization problems, this paper proposes a Multi-strategy Enhanced Social Network Search (MESNS) algorithm. The original SNS simulates human social behaviors through four decision-making emotions—imitation, conversation, disputation, and innovation—to perform population-based search. However, its uniform emotion selection mechanism and purely random interaction strategy may reduce convergence efficiency and weaken exploitation capability, particularly in the later stages of optimization. To overcome these limitations, MESNS incorporates three improvement strategies. First, an adaptive decision-making emotion selection mechanism is developed to dynamically adjust the probabilities of exploration and exploitation behaviors according to the iteration progress, thereby promoting a more symmetric and coordinated search transition over time. Second, an elite-guided communication strategy is introduced to enhance information propagation by integrating high-quality individuals into the interaction process, which improves convergence while maintaining population diversity. Third, a dynamic interaction radius adjustment mechanism is designed to adaptively regulate the search step size, achieving a better balance and dynamic symmetry between global exploration and local refinement. Extensive experiments are conducted on the IEEE CEC2014, CEC2017, and CEC2022 benchmark suites under multiple dimensional settings. The results demonstrate that MESNS achieves superior optimization accuracy, faster convergence speed, and improved solution stability compared with several state-of-the-art metaheuristic algorithms. Furthermore, the proposed algorithm is successfully applied to the three-dimensional wireless sensor network deployment optimization problem, where it produces a more uniformly distributed and spatially balanced sensor layout, reduces coverage holes and redundant overlaps, and thus exhibits desirable symmetry in deployment structure and sensing coverage. These findings indicate that MESNS is an effective and competitive optimization framework for complex global optimization tasks with both theoretical significance and practical value from the perspective of symmetry.

Keywords: social network search; metaheuristic algorithm; global optimization; adaptive mechanism; wireless sensor network deployment

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1. Introduction

With the rapid development of science and technology, complex optimization problems are widely found in various fields such as engineering design [1], intelligent manufacturing [2], communication systems [3], energy dispatching [4], and data analysis [5]. These problems typically exhibit characteristics such as high dimensionality, strong nonlinearity, multimodality, and complex constraints, resulting in a highly irregular and nonconvex search space. In many practical scenarios, the search space may also exhibit various forms of symmetry or asymmetry, such as symmetric variable interactions, balanced structural distributions, or symmetric constraints among decision variables. Effectively exploiting such symmetry information can help reduce redundant searches, improve search efficiency, and enhance solution quality. However, the coexistence of symmetric and asymmetric features often further increases the complexity of the optimization process. In practical applications, optimization objectives often conflict with each other, while computational efficiency and solution stability must also be considered, further increasing the difficulty of solving the problem. Traditional optimization methods based on gradients or analytical models usually rely on assumptions of continuity and differentiability and are sensitive to initial conditions, easily getting trapped in local optima or incurring excessive computational costs when facing complex search spaces. Moreover, these methods are generally less effective in handling optimization problems with complex symmetric structures, irregular landscapes, or multiple local optima. Therefore, obtaining high-quality approximate optimal solutions under limited computational resources has become one of the key challenges in modern optimization theory and engineering application research.

Against this backdrop, intelligent optimization algorithms, due to their low dependence on problem structure, wide applicability, and excellent global search capabilities, have gradually become important tools for solving complex optimization problems [6]. In particular, swarm intelligence optimization algorithms, inspired by the collective behavior of groups in nature, achieve efficient exploration of the solution space by simulating cooperation, competition, and information sharing mechanisms among individuals. Typical examples include traditional particle swarm optimization [7], genetic algorithms [8], and gray wolf optimization [9], as well as more recently proposed algorithms such as the Love Evolution Algorithm [10], Graduate Student Evolution Algorithm [11], and Hawk Fish Optimization Algorithm [12]. These algorithms have been widely applied in continuous optimization, combinatorial optimization, and multi-objective optimization problems. However, with the continuous increase in problem size and complexity, and the NFL theorem indicating that no single algorithm performs well on all problems, existing swarm intelligence algorithms still reveal some common shortcomings in practical applications [13]. This motivates researchers to make targeted improvements to algorithms for the specific problems they need to solve, enabling the algorithms to perform well on their chosen problems. Table 1 details the advantages and limitations of wireless sensor deployment methods.

Table 1. Advantages and limitations of representative WSN deployment optimization methods.

Category	Representative Method	Main Idea	Advantages	Limitations	References
Deterministic deployment	Grid-based deployment	Nodes are placed according to a regular geometric pattern such as square or hexagonal grids	Simple implementation, low computational cost, easy to analyze theoretically	Difficult to adapt to irregular terrain and complex obstacles; poor flexibility in practical scenarios	[14]

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Deterministic deployment	Virtual force algorithm	Nodes move according to attraction and repulsion forces among sensors and targets	Can improve coverage and connectivity simultaneously	Easily trapped in local optima; sensitive to parameter settings	[15]
Computational geometry-based deployment	Voronoi-based deployment	Uses Voronoi partitioning to identify coverage holes and relocate nodes	Effective in detecting uncovered regions; improves coverage efficiency	Computationally expensive in large-scale networks; weak global search capability	[16]
Swarm intelligence-based deployment	Improved sand cat swarm optimization algorithm	Virtual force-directed improved sand cat swarm optimization algorithm	Fast convergence speed; easy implementation	May suffer from premature convergence and poor local exploitation in high-dimensional problems	[17]
Swarm intelligence-based deployment	Hybrid Marine predators optimization and improved particle swarm optimization	Aims to ensure cluster head selection and data transmission, thereby achieving efficient cluster head selection and data transmission.	Strong global exploration ability; suitable for discrete optimization	Slow convergence speed; may require many parameters	[18]
Swarm intelligence-based deployment	Adaptive binary particle swarm optimization	Optimization Using Particles	Good balance between exploration and exploitation; few parameters	Diversity decreases rapidly in later iterations; easily stagnates locally	[19]
Proposed method	MESNS	Combines adaptive emotion selection, elite-guided communication, and dynamic interaction radius	Better balance between exploration and exploitation; stronger robustness and convergence performance	Additional parameters need to be tuned	This paper

To address the medical feature selection problem, Mohammad H. Nadimi-Shahraki et al. proposed an enhanced whale optimization algorithm using a pooling mechanism and an effective search strategy [20]. Experiments show that the proposed algorithm is highly competitive and outperforms the comparison algorithms. To achieve energy-efficient cluster head selection in wireless sensor networks, Mandli Rami Reddy et al. proposed an energy-efficient cluster head selection method based on GWO [21]. This method utilizes the EECHIGWO algorithm to optimize cluster head selection, effectively improving the energy efficiency, average throughput, network stability, and network lifetime of WSNs. To plan the path of unmanned aerial vehicles (UAVs) and enable them to perform tasks safely and efficiently, Xie et al. proposed a multi-elite guided RIME optimization algorithm based on the RIME algorithm for UAV path planning in 3D space [22]. Experimental results show that the proposed algorithm achieves significantly improved performance after the improvements. To address the multi-cost optimization problem in cloud task scheduling, Qin et al. proposed an effective cloud task scheduling method based on RTH. Extensive experiments demonstrate that their improved strategy significantly enhances the original algorithm, proving its effectiveness and necessity [23]. To optimize the landing time window problem for manned spacecraft, Lu et al. proposed a quantum-enhanced whale optimization algorithm, which utilizes quantum computing technology to address these challenges. Experimental results show that QEWOA significantly outperforms traditional methods in terms of global search capability, optimization accuracy, and convergence speed, making it an ideal solution for optimizing the landing time window of low-altitude manned spacecraft [24]. To address the challenges of multi-UAV mobile edge computing and path planning, Yu et al., inspired by the behavior of bounty hunters, proposed the Bounty Hunter Optimizer. This algorithm employs a decentralized position update strategy based on local differences and stochastic perturbations, thereby effectively mitigating issues related to central bias and population collapse. Simulation tests conducted across various scales have demonstrated its superior performance [25]. To address real-world engineering problems and UAV path planning, Yu et al. enhanced the performance of the DBO algorithm through four distinct strategies, thereby improving exploration efficiency, exploitation effectiveness, and system stability. Across 52 benchmark functions, 19 engineering problems, and UAV path planning tasks, the proposed EDBO algorithm consistently outperformed existing state-of-the-art algorithms [26].

Social Network Search (SNS) is a population-based metaheuristic optimization algorithm proposed in recent years [27], inspired by the information dissemination and public opinion evolution behaviors in social network services. In SNS, each candidate solution is modeled as a "user" in the social network, and the search process is driven by simulated social interactions between users. By abstracting typical social behaviors, SNS defines four decision-making emotions—imitation, dialogue, debate, and innovation—to guide the generation and evolution of solutions. These emotions enable the algorithm to explore the search space from multiple perspectives, balancing information sharing, group influence, and stochastic creativity. With its flexible social interaction mechanism and simple structure, SNS has demonstrated strong competitiveness in solving numerical optimization problems and in practical engineering applications.

To address the economic load scheduling problem in large power systems, Mohamed H. Hassan proposed an enhanced social network search algorithm. This technique yields promising results for this complex challenge. Its ability to handle multiple constraints and its superior performance compared to other recent algorithms make it a valuable addition to the existing ELD (Economic Load Dispatch) solution toolset [28]. Based on the above research, this paper proposes a Multi-strategy Enhanced Social Network Search Algorithm (MESNS). This algorithm improves the original SNS through mechanisms such as Adaptive Decision-Making Emotion Selection, thereby enhancing the algorithm's ability to escape local optima and improving overall search efficiency and stability. The specific contributions of this paper are as follows:

1. A multi-strategy enhanced social network search algorithm (MESNS) is proposed to address the limitations of the original SNS in balancing exploration and exploitation, particularly its reduced efficiency in later iterations caused by equal-probability emotion selection and purely random interactions.
2. An adaptive decision-based emotion selection mechanism is designed to dynamically adjust behavioral tendencies during the search process, enabling a more effective trade-off between global exploration and local exploitation.
3. An elite-guided communication strategy is introduced to improve high-quality information dissemination among individuals, thereby accelerating convergence and enhancing solution accuracy.
4. A dynamic interaction radius regulation mechanism is developed to adaptively control the search step size, strengthening global exploration in early stages while promoting refined local exploitation in later stages.
5. Extensive experiments on IEEE CEC2014, CEC2017, and CEC2022 benchmark suites demonstrate the superior performance of the proposed MESNS in terms of optimization accuracy, convergence speed, and robustness. In addition, the algorithm is successfully applied to a three-dimensional wireless sensor network deployment problem, further validating its practical effectiveness.

The remainder of this paper is structured as follows. Section 2 briefly introduces the SNS algorithm. Section 3 describes the three strategies proposed in this study and presents the overall execution framework of the algorithm. Section 4 reports numerical experiments and performance comparisons on three benchmark test sets. In Section 5, the proposed MESNS algorithm is applied to the deployment problem of 3D wireless sensor networks, and its engineering effectiveness is analyzed. Finally, Section 6 concludes the paper.

2. Social Network Search

Social Network Search (SNS) is a metaheuristic optimization method inspired by human behavior. Its basic idea is to treat candidate solutions as "user opinions" within a social network. By simulating the process of users updating their opinions and increasing their popularity in social interactions, an iterative update strategy for solutions is constructed. SNS abstracts typical user behaviors in social networks into four "decision moods": Imitation, Conversation, Disputation, and Innovation. In each iteration, a mood is randomly selected for each individual to generate a new solution. Table 2 summarizes the main symbols and notations used throughout this paper to improve readability and consistency. Table 2 summarizes the symbols and notations used in this paper.

Table 2. Description of Main Symbols Used in MESNS.

Symbol	Description
N	Population size
D	Problem dimension
T	Maximum number of iterations
LB	Lower bound of the search space
UB	Upper bound of the search space
R	Interaction vector or search step
X_i	Position (solution vector) of the i-th individual
$f(X_i)$	Fitness value of the i-th individual
$Div(t)$	Population diversity at iteration t
M	Mean viewpoint of the selected discussion group
AF	Acceptance factor in disputation behavior
K	Number of elite individuals
P_e	Probability of selecting an interacting individual from the elite set

Let the optimization problem be finding the optimal solution within the feasible region $[LB, UB]$, with a network size of N_{user}. The viewpoint (solution vector) of the i-th user can be represented as:

$$X_i = [x_{i1}, x_{i2}, \dots, x_{iD}], \tag{1}$$

Its fitness value can be represented as $f_i = f(X_i)$, where D represents the dimension of the problem to be solved.

2.1. Four Decision-Making Emotions and Mathematical Models

2.1.1. Imitation

Imitation of emotions reflects the behavior of "users imitating others' opinions" in social networks: Individual i randomly selects another user j $(i \neq j)$, generates a new opinion around X_j, and combines the differences between X_i and X_j. Its update form can be represented as follows:

$$X_i^{new} = X_j + \text{rand}(-1,1) \times R, \tag{2}$$

where R represents the difference between the two viewpoints, which can be expressed as follows:

$$R = \text{rand}(0,1) \times (X_j - X_i), \tag{3}$$

where rand$(-1{,}1)$ and rand$(0{,}1)$ represent random vectors generated in intervals $[-1, 1]$ and $[0, 1]$, respectively. This sentiment achieves its search by introducing random perturbations around X_j, potentially generating developmental behavior close to 222 or exploratory behavior beyond the range of $[X_i, X_j]$.

2.1.2. Conversation

The "interaction sentiment" parameter describes the process by which users adjust their "viewpoints on a particular issue" after private chat discussions. The algorithm introduces a random "issue vector" X_k, which is shifted by an individual through interaction with a random user j. The update can be represented as follows:

$$X_i^{new} = X_k + R, \tag{4}$$

where R represents the offset, as shown below.

$$R = \text{rand}(0{,}1) \times \text{sign}(f_i - f_j) \times (X_j - X_i)., \tag{5}$$

where $\text{sign}(\cdot)$ represents the sign function, which determines the direction of movement of X_k by comparing X_i and X_j, and satisfies $i \neq j \neq k$. This emotion, by reflecting the directional modification of the issue along the direction of inter-individual differences, is biased towards local development.

2.1.3. Disputation

Debate and Emotional Simulation in Comment Sections/Group Discussions: Individuals are influenced by the average viewpoint of a group of users, while also considering the factor of "degree of persistence." Assuming a discussion group is randomly selected with N_r users, its mean can be expressed as follows:

$$M = \frac{1}{N_r}\sum_{t=1}^{N_r} X_t, \tag{6}$$

The acceptance factor is defined as follows:

$$AF = 1 + \text{round}(\text{rand}), \tag{7}$$

where $\text{rand} \in [0{,}1]$, $\text{round}(\cdot)$ represents the rounding function, making $AF \in \{1{,}2\}$. Therefore, the position update formula can be expressed as follows:

$$X_i^{new} = X_i + \text{rand}(0{,}1) \times (M - AF \times X_i), \tag{8}$$

This sentiment guides the search direction through the group average viewpoint M and uses AF to randomly adjust the step size, thus exhibiting different degrees of exploration/development characteristics at different stages.

2.1.4. Innovation

Innovative sentiment reflects users' "reconstruction" of a particular aspect of their viewpoints based on new ideas. The algorithm randomly selects dimension $d \in \{1{,}2, \dots, D\}$, and generates random new ideas for that dimension, which can be represented as follows:

$$n_{new}^d = lb_d + \text{rand}_1(ub_d - lb_d), \tag{9}$$

where $\text{rand}_1 \in [0{,}1]$, lb_d and ub_d are the upper and lower bounds of the d-th dimension. Then $t = \text{rand}_2(\text{rand}_2 \in [0{,}1])$ is interpolated with the current thought x_j^d of random user j, which can be specifically represented as follows:

$$x_{i,new}^d = t \times x_j^d + (1 - t) \times n_{new}^d, \tag{10}$$

This results in a new viewpoint vector, specifically represented as follows:

$$X_i^{new} = [x_{i1}, x_{i2}, \dots, x_{i,new}^d, \dots, x_{iD}]. \tag{11}$$

This emotion randomly reconstructs a single dimension, often exhibiting stronger leaps, and primarily serves the purpose of global exploration and escaping local optima.

2.2. Decision-Making Emotion Selection Mechanism

One characteristic of social networking services (SNS) is that in each iteration, each user performs only one of the four emotions mentioned above, and the selection is done in a uniformly random manner (the four emotions are triggered with equal probability). This mechanism is used to simulate the randomness and diversity of real-world social behavior over time, while avoiding search bias caused by fixed update strategies.

2.3. Network Rules

Because the optimization variables have upper and lower bounds, SNS truncates them dimension by dimension after generating X_i^{new}, which can be specifically represented as follows:

$$x_i = min(x_i, ub_i), x_i = max(x_i, lb_i). \tag{12}$$

where x_i represents the corresponding dimension component in the new solution, and ub_i, lb_i are the upper and lower bounds of that dimension, respectively.

2.4. Publishing Rule

The social networking service (SNS) employs a fitness-based "posting rule": if a new viewpoint is superior to the current one, it is accepted and replaced; otherwise, the original viewpoint is retained. Specifically, this can be expressed as follows:

$$\text{for minimization problem:} X_i = \begin{cases} X_i, f(X_i) < f(X_i^{new}) \\ X_i^{new}, \quad f(X_i^{new}) \geq f(X_i) \end{cases} \tag{13}$$

2.5. Initialization and Algorithm Flow

2.5.1. Initialization

SNS first generates an initial network (initial population). For each user, their opinion vector is randomly initialized, which can be represented as follows:

$$X_0 = LB + \text{rand}(0,1) \times (UB - LB). \tag{14}$$

Then, the fitness of each user was calculated.

2.5.2. Iterative Updates (Popularity Enhancement Phase)

Repeat for each iteration, for each user:

1. Select an emotion $m \in \{1,2,3,4\}$ in a uniformly random manner, and generate X_i^{new} according to the corresponding formula.
2. Boundary constraints
3. Calculate fitness value
4. Update X_i according to release/replacement policy

2.5.3. Termination Conditions

The algorithm can use the maximum number of iterations, reaching a given precision threshold, or the optimal value no longer improving after a certain number of iterations as termination conditions. After termination, the algorithm outputs the optimal solution of the entire network as the final result.

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3. Multi-Strategy Enhanced Social Network Search Algorithm (MESNS)

To further enhance the convergence performance, search efficiency, and robustness of the original Social Network Search (SNS) algorithm, a multi-strategy enhanced variant named MESNS is proposed in this paper. The proposed MESNS incorporates adaptive communication strategies and elite-guided information propagation while preserving the original social behavior modeling framework. The main improvements of ISNS include:

1. an adaptive decision-making emotion selection mechanism.
2. an elite-guided communication strategy.
3. a dynamic interaction radius for step-size control.

These mechanisms aim to achieve a better balance between global exploration and local exploitation during different stages of the optimization process.

3.1. Adaptive Decision-Making Emotion Selection

In the original SNS algorithm, the four decision-making emotions—imitation, conversation, disputation, and innovation—are triggered with equal probability in each iteration. Although this strategy maintains population diversity, it may lead to inefficient search behavior in later iterations.

To address this issue, MESNS introduces an adaptive emotion selection mechanism that dynamically adjusts the activation probabilities of different emotions according to the iteration progress. Let T denote the maximum number of iterations and t the current iteration index. The probabilities of selecting each emotion are defined as:

$$\begin{gathered} P_{sum} = \alpha\left(1-\frac{t}{T}\right)+\beta\left(1-\frac{t}{T}\right)+\gamma\frac{t}{T}+\delta\frac{t}{T} \\ P_{\text{imitation}}(t) = \alpha\left(1-\frac{t}{T}\right)/P_{sum}, \\ P_{\text{innovation}}(t) = \beta\left(1-\frac{t}{T}\right)/P_{sum}, \\ P_{\text{conversation}}(t) = \gamma\frac{t}{T}/P_{sum}, \\ P_{\text{disputation}}(t) = \delta\frac{t}{T}/P_{sum}, \end{gathered} \tag{15}$$

where $\alpha, \beta, \gamma, \delta \in (0,1)$ and satisfy $\alpha+\beta+\gamma+\delta = 1$. This mechanism encourages global exploration in the early search stage through imitation and innovation behaviors, while gradually shifting toward local exploitation via conversation and disputation behaviors in later iterations.

The linear transition mechanism in Equation (15) is designed to provide a smooth and stable balance between exploration and exploitation throughout the optimization process. In the early search stage, larger probabilities are assigned to imitation and innovation behaviors to encourage population diversity and global exploration. As the iteration proceeds, the probabilities of conversation and disputation gradually increase, promoting local exploitation and solution refinement.

A linear scheduling strategy is adopted because of its simplicity, stability, and wide use in many metaheuristic algorithms, such as the inertia weight adjustment in particle swarm optimization and the control parameter reduction in whale optimization and gray wolf optimization. Compared with nonlinear or abrupt transition schemes, the linear mechanism avoids excessive oscillation in behavioral selection probabilities and provides a smoother evolution of the search dynamics.

From the perspective of optimization dynamics, Equation (15) does not provide a strict theoretical guarantee of global convergence, which is a common limitation of most population-based metaheuristic algorithms. However, the gradual reduction in exploratory behaviors and the increasing emphasis on exploitative behaviors help improve search stability in later iterations and reduce the probability of random divergence. Therefore, the proposed mechanism should be regarded as an empirically motivated yet practically effective strategy for maintaining a proper exploration–exploitation balance.

3.2. Elite-Guided Communication Strategy

In the original SNS framework, users interact with randomly selected individuals, which may introduce low-quality information into the updating process. Furthermore, to address the multi-sensor collaborative observation scheduling problem in space surveillance networks, Long et al. proposed an improved particle swarm optimization algorithm combining back learning, neighborhood adjustment, and particle perturbation mechanisms. The algorithm comprehensively considers various constraints such as observation time, observation frequency, and sensor capabilities for each on-orbit space object in its modeling. Experimental results show that, compared with several other effective algorithms, the proposed algorithm improves task completion rates by approximately 17.01%, 19.1%, 13.53%, and 13.99%, respectively; while reducing resource consumption rates by approximately 55.48% and 36.38%, respectively [29]. To improve the quality of information dissemination, MESNS incorporates an elite-guided communication strategy.

At each iteration, the top $K = 0.1N$ users with the best fitness values are selected as the elite set [22]:

$$\mathcal{E} = \{X_{(1)}, X_{(2)}, \dots, X_{(K)}\}, \tag{16}$$

where $X_{(K)}$ denotes the k-th best solution in the current population. During imitation and conversation behaviors, the interacting user j is selected from the elite set ε with probability $p_e = 0.3$, and from the entire population with probability $1 - p_e$. This hybrid selection strategy can be expressed as [30]:

$$X_j = \begin{cases} X_{\text{elite}}, & \text{with probability} p_e, \\ X_{\text{random}}, & \text{with probability} 1 - p_e. \end{cases} \tag{17}$$

By emphasizing elite individuals while retaining randomness, the algorithm accelerates convergence without significantly increasing the risk of premature stagnation.

3.3. Dynamic Interaction Radius Adjustment

To further improve search adaptability, MESNS introduces a dynamic interaction radius to regulate the step size of social interactions. Specifically, the interaction vector R is redefined as:

$$R = \lambda(t) \cdot rand(0,1) \cdot (X_j - X_i), \tag{18}$$

where $\lambda(t)$ is a time-varying control factor defined as:

$$\lambda(t) = \lambda_{max} - (\lambda_{max} - \lambda_{min}) \frac{t}{T}, \tag{19}$$

With $\lambda_{max} = 1$ and $\lambda_{min} = 0.1$ denoting the maximum and minimum interaction radii, respectively [25].

A linear decay strategy is adopted for the interaction radius because it provides a simple and stable transition from global exploration to local exploitation. In the early stage of the search, a relatively large interaction radius encourages individuals to explore distant regions of the search space. As the iteration proceeds, the interaction radius gradually decreases, enabling more refined local exploitation around promising solutions.

Compared with nonlinear decay or adaptive feedback-based strategies, the linear model has two main advantages. First, it introduces fewer control parameters and therefore reduces the risk of overfitting the algorithm to specific optimization problems.

Second, the linear reduction process is smooth and predictable, which helps avoid abrupt changes in search behavior and improves the stability of the optimization process.

Although more sophisticated schemes, such as exponential decay, cosine scheduling, or population-diversity-based adaptive adjustment, may further improve performance in some scenarios, they usually require additional parameter tuning and increase computational complexity. Therefore, the linear decay mechanism is adopted in this work as a compromise between effectiveness, simplicity, and robustness.

This strategy enables large exploratory moves at early iterations and fine-grained local search in later stages, thus improving convergence stability and solution accuracy.

The three proposed mechanisms are designed to work cooperatively during the optimization process. The adaptive emotion selection mechanism dynamically adjusts the balance between exploration and exploitation. The elite-guided communication strategy improves the dissemination of high-quality information and accelerates convergence. Meanwhile, the dynamic interaction radius adaptively controls the search step size to maintain exploration ability in the early stage and strengthen local refinement in the later stage. Through the cooperation of these three mechanisms, MESNS can achieve better convergence speed, solution accuracy, and robustness than the original SNS.

3.4. Overall Algorithm Flow of MESNS

The overall procedure of the proposed MESNS algorithm is summarized as follows:

1. Initialize the population randomly within the feasible search space.
2. Evaluate the fitness value of each user.
    (a) For each iteration $t = 1,2,3,\dots,T$:
    (b) For each user i:
        i. Select a decision-making emotion according to the adaptive probability model.
        ii. Generate a new candidate solution according to the selected decision-making emotion. For imitation and conversation behaviors, the interacting individual is selected from the elite set with probability (p_e), otherwise it is randomly selected from the population. In addition, the interaction vector is scaled by the dynamic interaction radius ($\lambda(t)$) to adaptively adjust the search step size during different stages of the optimization process. For disputation and innovation behaviors, the original SNS updating rules are retained.
        iii. Apply boundary constraints to the new solution.
        iv. Evaluate the fitness value.
        v. Update the user's viewpoint based on the publishing rule.
3. Output the best solution obtained as the final result.

By integrating adaptive emotion selection, elite-guided communication, and dynamic interaction radius control, MESNS significantly enhances the balance between exploration and exploitation. These improvements strengthen the algorithm's convergence speed, solution accuracy, and robustness when solving complex numerical optimization problems. Figure 1 shows the algorithm flowchart of the MESNS proposed in this paper.

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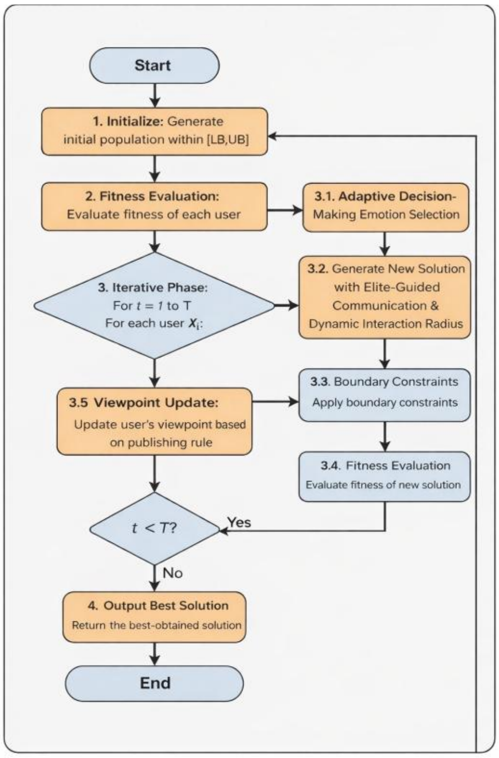

Figure 1. MESNS Algorithm Flowchart.

3.5. Time Complexity Analysis

Let N denote the population size, D the problem dimension, T the maximum number of iterations, and C the computational cost of one fitness evaluation. During initialization, MESNS randomly generates N candidate solutions in a D-dimensional search space and evaluates their fitness values, resulting in a complexity of $O(ND + NC)$. In each iteration, for each individual, the adaptive emotion selection mechanism only introduces constant-time overhead, i.e., $O(1)$. The four updating behaviors, namely imitation, conversation, disputation, and innovation, all involve vector operations over D dimensions, and thus require $O(D)$ time in general. The boundary handling step also costs $O(D)$, while the greedy replacement rule takes $O(1)$. In addition, fitness evaluation requires $O(C)$. Therefore, the main per-iteration cost of population updating is $O(N(D + C))$. Compared with the original SNS, the additional computational burden of MESNS mainly comes from the elite-guided communication strategy, where the top-K individuals are selected to form an elite set. If sorting is used, this process requires $O(NlogN)$ time per iteration. Hence, the overall time complexity of MESNS can be expressed as $O(T[N(D + C) + NlogN])$. When K and the discussion group size are treated as constants, MESNS preserves the same dominant complexity order as the original SNS, namely $O(TN(D + C))$, indicating that the proposed enhancement strategies improve search performance without significantly increasing the asymptotic computational complexity.

From the perspective of computational scalability, the additional mechanisms introduced in MESNS mainly include adaptive emotion selection, elite-guided communication, and dynamic interaction radius adjustment. Among them, the main additional computational cost comes from elite selection, which requires sorting the population and has a complexity of $(O(N\ log\ N))$. However, this additional cost does not change the dominant complexity order of the algorithm. Therefore, the overall time complexity of MESNS remains $(O(TN(D + C)))$, which is of the same order as the original SNS. This indicates that MESNS can be extended to larger-scale optimization problems without causing a significant increase in computational burden.

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4. Numerical Experiments

In this section, we present numerical experiments on the proposed MESNS algorithm. To ensure reproducibility, we first detail the parameter settings for each comparison algorithm, and then conduct a thorough experimental analysis on the CEC2014 [31], CEC2017 [32] and CEC2022 [33] test sets. To determine whether there are significant differences between the algorithms, we performed a Friedman rank test, the details of which are as follows.

4.1. Competitor Algorithms and Parameters Setting

This section details the parameter settings for all comparison algorithms to improve experimental reproducibility. The comparison algorithms include Snow Ablation Optimizer (SAO) [34], Secretary Bird Optimization Algorithm (SBOA) [35], Escape Algorithm (ESC) [36], Gold Rush Optimizer (GRO) [37], Connected Banking System Optimizer (CBSO) [38], Teaching-Learning-Based Optimization (TLBO) [39], Artificial Lemming Algorithm (ALA) [40], Stellar Oscillation Optimizer (SOO) [41], Phototropic growth algorithm (PGA) [42], Snow Geese Algorithm (SGA) [43], and Social Network Search (SNS). The parameter configurations and experimental settings for all algorithms are summarized in Table 3.

Table 3. Compare algorithm parameter settings.

Algorithms	Parameters	Value
SAO	*k*	1
SBOA	*beta*	1.5
ESC	*eliteSize, beta_base*	5, 1.5
GRO	*sigma_initial*	2
CBSO	*beta*	1.5
TLBO	--	--
ALA	*beta*	1.5
SOO	--	--
PGA	--	--
SGA	*T*	1
SNS	--	--

4.2. Numerical Results and Analysis on the CEC2014 Benchmark

To verify the performance of the proposed MESNS, the IEEE CEC2014 benchmark suite was used as the evaluation platform in this section. Two dimensional scenarios were designed, namely 30-dimensional and 50-dimensional, and comparative experiments were conducted with 11 other classic metaheuristic algorithms. To systematically evaluate the global exploration and local exploitation capabilities of MESNS, the focus was on three core performance indicators: optimization accuracy, convergence speed, and result stability. The experimental results are shown below. Tables 4 and 5 summarize the mean and standard deviation of each algorithm after 30 runs, where ave represents the mean of 30 runs and std represents the standard deviation of 30 runs. Figure 2 shows the convergence curves of each algorithm on the test set, and Figure 3 shows the box plots of the results of 30 runs of each algorithm. The optimal experimental results for each function are shown in bold.

In the 30-dimensional and 50-dimensional tests of the CEC2014 benchmark functions, to comprehensively evaluate the algorithm's performance, all functions were run independently 30 times, and their mean (ave) and standard deviation (std) were calculated. Tables 4 and 5 show that the proposed MESNS algorithm achieved superior or

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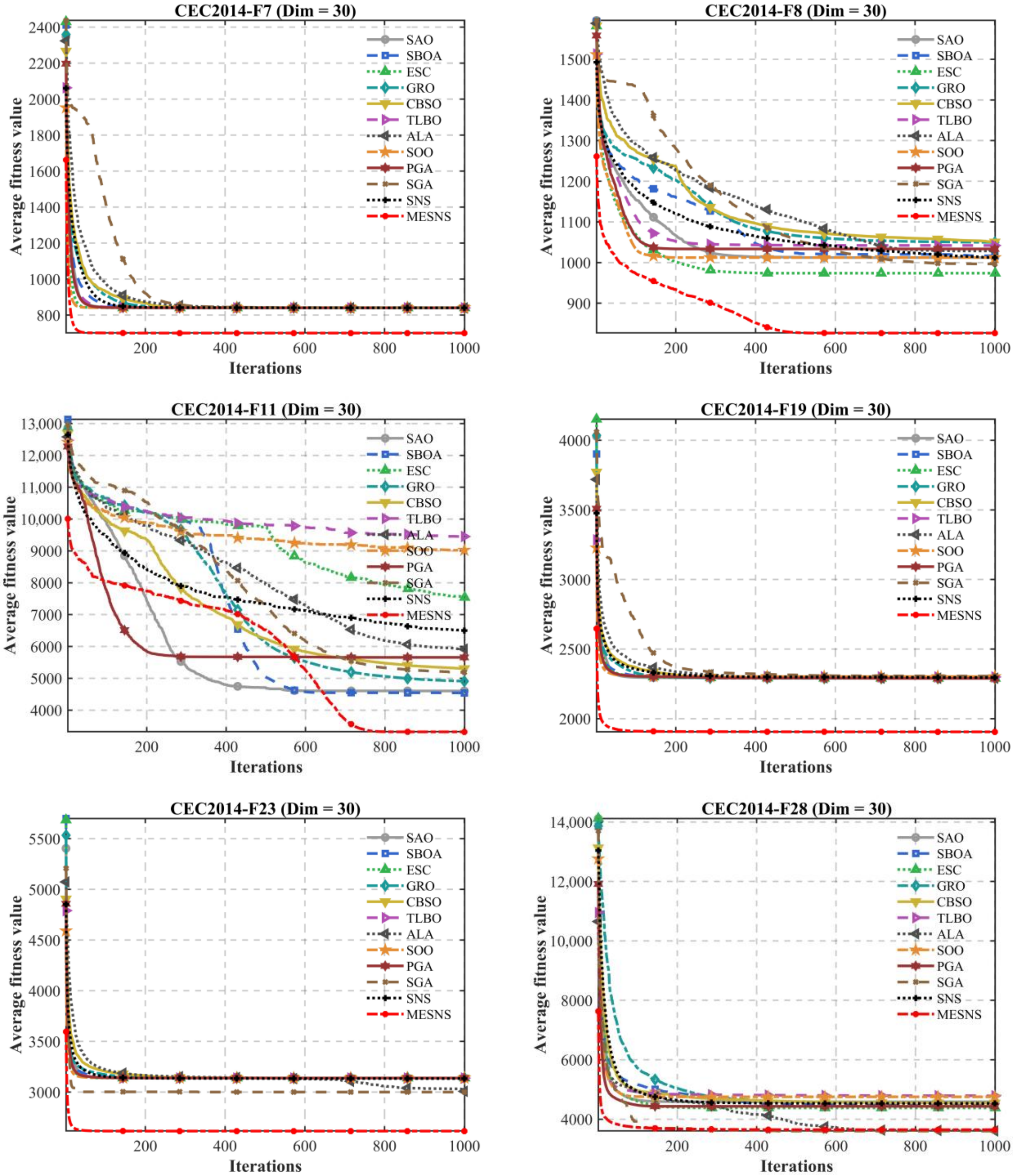
significantly superior results on most test functions. In the 30-dimensional case, MESNS performed exceptionally well on typical unimodal and multimodal functions such as F1, F2, F3, and F4. Not only were its mean values significantly better than other comparative algorithms, but its standard deviation was also generally smaller, indicating that it possesses good stability and robustness while maintaining high solution accuracy. Especially on some complex mixed functions (such as F16–F18), MESNS still maintained a low mean level and small fluctuation amplitude, demonstrating that the algorithm has a strong balance mechanism between global exploration capability and local exploitation capability in complex search spaces. When the problem dimension increased to 50, the overall performance of all algorithms declined. However, MESNS still maintained leading or competitive results on most functions, with a relatively small increase in the average value and the standard deviation remaining within a reasonable range, demonstrating good scalability and adaptability to high dimensions. In summary, MESNS exhibited superior optimization performance and stability in the CEC2014 benchmark tests across different dimensions, validating the effectiveness and reliability of the proposed strategy in complex high-dimensional optimization problems.

Figure 2 shows the convergence curves, which provide a more intuitive comparison of the optimization processes of different algorithms on the CEC2014 test function. Overall, MESNS exhibits faster convergence speed and better final convergence accuracy on most functions. In the early stages of iteration, the MESNS curve shows a significant decrease, rapidly reducing the objective function value and demonstrating strong global search capabilities. In the mid-to-late stages, its convergence curve remains stable and continues to decline without significant oscillations or premature stagnation, indicating that the algorithm possesses good local exploitation and solution refinement capabilities. In contrast, some of the compared algorithms, while showing a certain downward trend in the early stages, tend to plateau or experience a significant slowdown in convergence speed, reflecting their shortcomings in escaping local optima or maintaining population diversity. Furthermore, on complex multimodal and mixed functions, the MESNS convergence curve is smoother overall with smaller fluctuations, further verifying the stability and robustness of its search process. As the problem dimension increases, the convergence speed of each algorithm decreases, but MESNS still maintains a relatively fast convergence trend and high solution quality, indicating that it has good scalability and comprehensive performance advantages in high-dimensional optimization problems.

The performance distribution results shown in Figure 3 provide a more comprehensive reflection of the overall performance differences in different algorithms on the CEC2014 test functions. As can be seen from the figure, MESNS's performance distribution is more concentrated and its overall position is better on most test functions, with its results generally falling in the lower range, indicating that the algorithm can stably obtain high-quality solutions under different function types. Furthermore, compared to other comparative algorithms, MESNS's distribution range is more compact and its dispersion is smaller, indicating lower fluctuations and stronger robustness. In contrast, the performance distribution of some algorithms shows a large span, even exhibiting a significant long-tail phenomenon, reflecting their susceptibility to performance degradation or instability on some complex functions. Moreover, in multimodal and mixed function categories, MESNS maintains a better distribution position and smaller dispersion, demonstrating its adaptability and stability advantages in complex search spaces. Overall, the performance distribution characteristics shown in Figure 3 further verify the comprehensive advantages of MESNS in terms of overall solution quality, consistency, and robustness, which corroborates the aforementioned statistical results and convergence curve analysis.

These results indicate that MESNS not only performs well on low-dimensional problems but also maintains strong optimization capability as the problem dimension increases, demonstrating good scalability for high-dimensional optimization tasks.



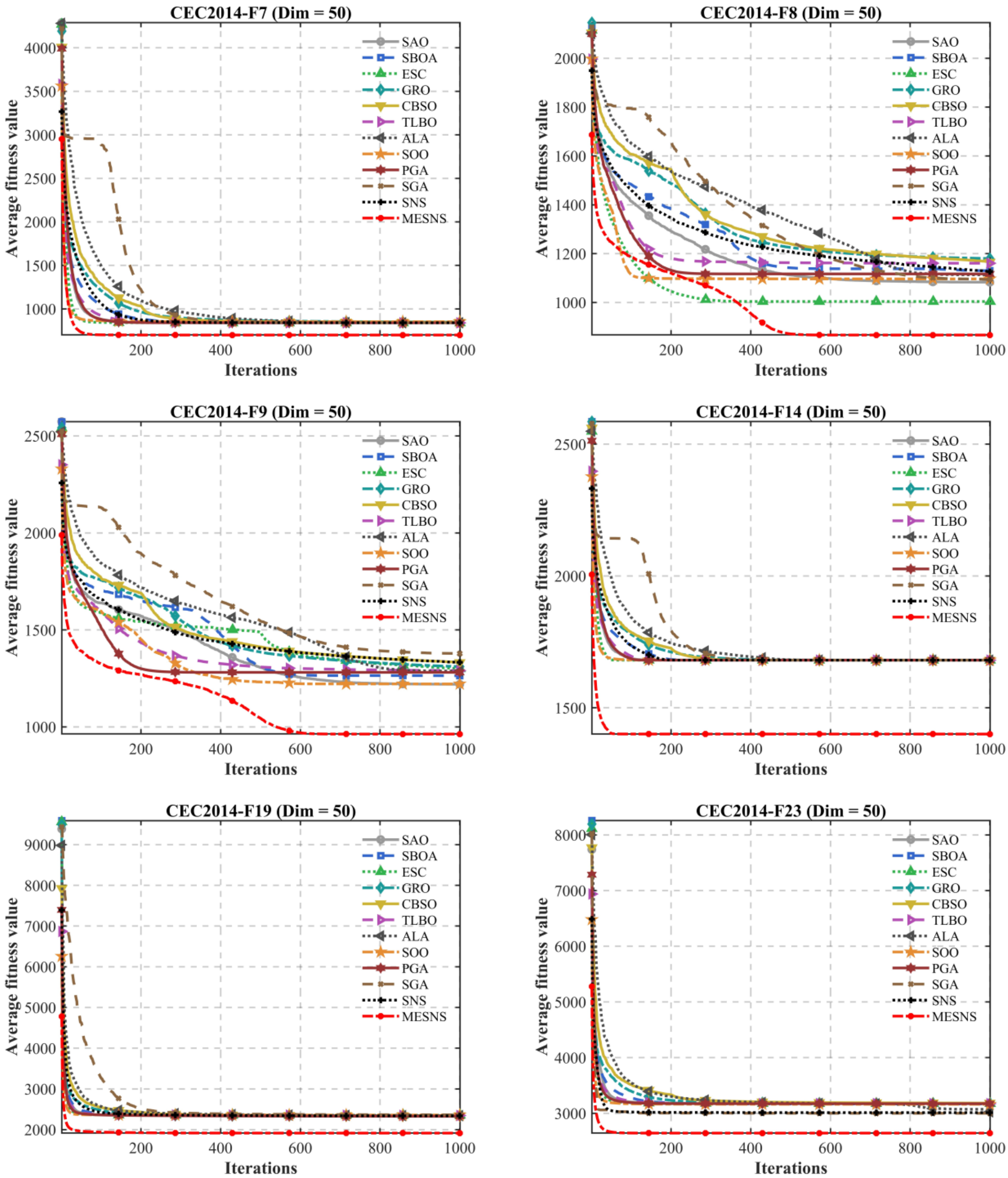


Figure 2. Convergence Speed Comparison of Different Algorithms on the CEC2014 Test Functions.

Table 4. Performance Comparison of the Proposed Algorithms on the 30-Dimensional CEC2014 Benchmark.

ID	Metric	SAO	SBOA	ESC	GRO	CBSO	TLBO	ALA	SOO	PGA	SGA	SNS	MESNS
F1	ave	1.8145E+06	4.7240E+06	9.0778E+06	1.1356E+07	5.7329E+05	7.4692E+05	4.0391E+06	2.3262E+06	5.3178E+06	6.2424E+07	7.4340E+06	1.1488E+03
	std	1.0835E+06	2.5677E+06	5.3287E+06	5.2649E+06	4.7757E+05	4.6110E+05	2.4513E+06	1.7019E+06	2.5087E+06	2.4376E+07	3.8717E+06	3.4271E+02
F2	ave	1.2740E+04	6.3155E+03	1.0055E+04	2.4167E+06	5.9051E+06	**5.8706E+02**	9.9138E+03	3.0525E+06	1.3621E+04	3.9401E+08	1.6724E+04	**2.0000E+02**
	std	1.0357E+04	7.2539E+03	6.5680E+03	4.2056E+06	1.5952E+07	**4.0656E+02**	7.8925E+03	8.2891E+06	1.2620E+04	3.6893E+08	9.5261E+03	**1.3910E-05**
F3	ave	9.8939E+03	3.4928E+03	2.4588E+03	1.3617E+04	6.1311E+02	6.4008E+03	1.2296E+03	3.0144E+02	6.2721E+03	5.3093E+04	3.0506E+03	**3.0000E+02**
	std	7.4375E+03	2.5943E+03	3.8349E+03	5.8247E+03	2.9590E+02	2.8716E+03	6.0949E+02	2.7957E+00	5.7584E+03	9.7528E+03	1.4677E+03	**7.0787E-04**
F4	ave	4.9085E+02	4.9303E+02	5.0623E+02	5.1933E+02	5.1295E+02	4.8644E+02	5.2132E+02	5.5863E+02	5.0682E+02	6.7326E+02	5.2421E+02	**4.0214E+02**
	std	3.1196E+01	2.4871E+01	3.1635E+01	4.3321E+01	2.4930E+01	3.7371E+01	4.6097E+01	3.2816E+01	3.2085E+01	1.0475E+02	3.7161E+01	**1.4734E+00**
F5	ave	5.2095E+02	5.2013E+02	5.2095E+02	5.2017E+02	5.2035E+02	5.2097E+02	5.2073E+02	5.2097E+02	5.2012E+02	5.2042E+02	5.2057E+02	**5.2004E+02**
	std	1.4620E-01	9.7175E-02	5.4943E-02	6.3586E-02	1.2843E-01	6.4893E-02	1.7778E-01	5.3775E-02	1.1401E-01	1.9458E-01	3.3837E-02	**1.6612E-02**
F6	ave	6.0523E+02	6.0954E+02	**6.0299E+02**	6.1178E+02	6.2096E+02	6.1492E+02	6.1899E+02	6.0865E+02	6.1352E+02	6.3394E+02	6.1255E+02	6.0368E+02
	std	2.6019E+00	2.5162E+00	1.4576E+00	2.0887E+00	2.9419E+00	2.1486E+00	3.6213E+00	2.0645E+00	3.8730E+00	2.8610E+00	2.0352E+00	**9.7573E-01**
F7	ave	7.0001E+02	7.0001E+02	7.0000E+02	7.0065E+02	7.0092E+02	7.0004E+02	7.0001E+02	7.0075E+02	7.0001E+02	7.0824E+02	7.0006E+02	**7.0000E+02**
	std	1.1059E-02	1.7087E-02	5.0077E-03	2.7049E-01	1.9457E-01	1.1396E-01	1.2566E-02	3.8056E-01	1.2347E-02	7.8108E+00	3.6689E-02	**8.5476E-11**
F8	ave	8.4389E+02	8.5172E+02	**8.1101E+02**	8.6960E+02	8.7113E+02	8.6862E+02	8.6251E+02	8.3968E+02	8.5416E+02	9.6864E+02	8.4241E+02	8.2837E+02
	std	1.3659E+01	1.8859E+01	3.1780E+00	1.4899E+01	1.7158E+01	1.4798E+01	1.6764E+01	1.0764E+01	1.6165E+01	2.8437E+01	7.4093E+00	**2.2373E+00**
F9	ave	9.5368E+02	9.7080E+02	9.6900E+02	9.8358E+02	9.9121E+02	9.7407E+02	9.8921E+02	9.5566E+02	9.7606E+02	1.0933E+03	9.9321E+02	**9.2607E+02**
	std	1.5488E+01	2.0168E+01	2.0866E+01	2.0760E+01	1.7364E+01	1.5493E+01	2.1412E+01	3.2960E+01	2.5488E+01	3.5013E+01	2.3967E+01	**3.7548E+00**
F10	ave	2.9414E+03	1.7686E+03	**1.2559E+03**	2.9735E+03	2.8606E+03	3.0305E+03	3.5804E+03	1.6530E+03	3.0897E+03	5.3311E+03	1.9486E+03	2.3240E+03
	std	6.8125E+02	3.5035E+02	1.9273E+02	4.2441E+02	6.0578E+02	7.7174E+02	6.7105E+02	2.5652E+02	7.5888E+02	7.7107E+02	**1.4330E+02**	2.0277E+02
F11	ave	3.7885E+03	3.8492E+03	6.3952E+03	3.9304E+03	4.3046E+03	7.8252E+03	5.0483E+03	7.2110E+03	4.5254E+03	6.0296E+03	5.3838E+03	**3.2767E+03**
	std	5.8548E+02	7.8017E+02	4.4934E+02	5.9617E+02	5.3898E+02	5.1364E+02	5.4300E+02	9.8605E+02	6.4269E+02	1.0506E+03	3.5612E+02	**2.3099E+02**
F12	ave	1.2013E+03	**1.2002E+03**	1.2019E+03	1.2005E+03	1.2004E+03	1.2028E+03	1.2013E+03	1.2024E+03	1.2004E+03	1.2017E+03	1.2010E+03	1.2003E+03
	std	1.1787E+00	1.1001E-01	3.7460E-01	2.2991E-01	1.7277E-01	2.8048E-01	4.4744E-01	3.5264E-01	2.6671E-01	4.1927E-01	1.2396E-01	**5.7030E-02**
F13	ave	1.3005E+03	1.3004E+03	1.3003E+03	1.3003E+03	1.3003E+03	1.3004E+03	1.3003E+03	1.3004E+03	1.3005E+03	1.3005E+03	1.3004E+03	**1.3002E+03**
	std	8.4763E-02	8.9126E-02	4.1483E-02	6.3695E-02	5.6486E-02	7.6196E-02	7.3829E-02	8.5392E-02	1.0540E-01	1.3316E-01	9.7528E-02	**2.5376E-02**
F14	ave	1.4005E+03	1.4003E+03	1.4003E+03	1.4003E+03	1.4003E+03	1.4003E+03	1.4003E+03	1.4004E+03	1.4006E+03	1.4005E+03	1.4003E+03	**1.4002E+03**
	std	1.9053E-01	7.5785E-02	4.6597E-02	4.4144E-02	3.7302E-02	3.7419E-02	1.5493E-01	1.3035E-01	2.5637E-01	3.7573E-01	5.2154E-02	**2.0693E-02**
F15	ave	1.5163E+03	1.5085E+03	1.5099E+03	1.5104E+03	1.5099E+03	1.5180E+03	1.5104E+03	1.5195E+03	1.5059E+03	1.6925E+03	1.5142E+03	**1.5028E+03**
	std	3.9301E+00	3.0364E+00	9.1229E-01	6.6779E+00	2.1800E+00	6.2316E+00	2.4300E+00	1.4281E+01	2.0030E+00	2.3534E+02	5.1545E+00	**4.0397E-01**
F16	ave	1.6126E+03	1.6109E+03	1.6117E+03	1.6111E+03	1.6118E+03	1.6122E+03	1.6124E+03	1.6117E+03	1.6125E+03	1.6129E+03	1.6113E+03	**1.6102E+03**

	std	4.2303E-01	8.8304E-01	4.2744E-01	6.1282E-01	3.4123E-01	**3.3773E-01**	4.2094E-01	3.9570E-01	4.9060E-01	4.0966E-01	3.8715E-01	3.6937E-01
F17	ave	1.9503E+05	4.4126E+05	1.0970E+06	4.1958E+05	3.0743E+03	3.9490E+05	6.9056E+03	5.6742E+05	6.6068E+05	2.7974E+06	4.0343E+05	**2.1486E+03**
	std	1.1293E+05	2.8204E+05	7.5281E+05	3.5430E+05	3.7996E+02	3.0833E+05	1.8877E+03	4.6762E+05	5.0401E+05	1.8576E+06	3.8555E+05	**8.5147E+01**
F18	ave	4.4644E+03	5.1453E+03	4.2189E+03	3.0035E+03	1.9558E+03	3.7504E+03	6.0075E+03	6.8591E+03	8.2803E+03	6.8402E+04	2.7710E+03	**1.8359E+03**
	std	6.3875E+03	4.9720E+03	2.6815E+03	1.3679E+03	3.4414E+01	2.1439E+03	8.7296E+03	6.8226E+03	7.1243E+03	2.0659E+05	1.1936E+03	**6.1686E+00**
F19	ave	1.9118E+03	1.9070E+03	1.9080E+03	1.9102E+03	1.9097E+03	1.9122E+03	1.9109E+03	1.9151E+03	1.9076E+03	1.9437E+03	1.9170E+03	**1.9050E+03**
	std	1.7512E+01	9.9363E-01	1.2181E+01	1.1164E+01	1.3998E+00	1.0278E+01	1.3205E+01	1.6075E+01	1.4565E+00	2.5734E+01	2.6552E+01	**7.3317E-01**
F20	ave	3.0208E+04	4.5320E+03	8.0799E+03	1.5267E+04	2.0983E+03	6.0770E+03	2.2216E+03	2.2110E+03	1.4728E+04	2.8933E+04	9.9143E+03	**2.0243E+03**
	std	1.8069E+04	2.5723E+03	3.8402E+03	4.6382E+03	2.4602E+01	3.0007E+03	5.8918E+01	6.9278E+01	5.0831E+03	9.1285E+03	3.8371E+03	**2.7664E+00**
F21	ave	1.4002E+05	9.5093E+04	3.4694E+05	1.2457E+05	3.0896E+03	1.3207E+05	4.8400E+03	5.4909E+04	3.3667E+05	6.5181E+05	8.1775E+04	**2.5274E+03**
	std	1.3592E+05	7.7721E+04	2.4797E+05	1.0890E+05	3.8919E+02	9.8781E+04	7.3858E+02	4.1468E+04	3.5059E+05	5.6206E+05	9.3971E+04	**7.1889E+01**
F22	ave	2.6482E+03	2.3870E+03	2.3854E+03	2.4233E+03	2.4764E+03	2.4896E+03	2.5212E+03	2.4803E+03	2.5641E+03	3.0961E+03	2.4960E+03	**2.2496E+03**
	std	2.2809E+02	1.1109E+02	1.0748E+02	8.8444E+01	1.0060E+02	1.1620E+02	2.2554E+02	1.1896E+02	1.6518E+02	2.5786E+02	1.2859E+02	**6.9306E+00**
F23	ave	2.6152E+03	2.6152E+03	2.6152E+03	2.6155E+03	2.6153E+03	2.6152E+03	**2.5047E+03**	2.6163E+03	2.6152E+03	2.5356E+03	2.6076E+03	2.6152E+03
	std	1.3377E-05	1.9458E-06	3.8166E-04	2.4697E-01	8.6931E-02	5.7781E-12	2.8023E+00	1.9486E+00	5.0447E-03	5.9662E+01	2.9240E+01	**1.9256E-12**
F24	ave	2.6090E+03	2.6000E+03	2.6292E+03	2.6000E+03	2.6109E+03	2.6000E+03	2.6001E+03	**2.6000E+03**	2.6338E+03	2.6003E+03	2.6000E+03	2.6210E+03
	std	1.0877E+01	4.0022E-04	5.5270E+00	5.7612E-03	1.2528E+01	3.4159E-03	1.0047E-01	**3.9437E-05**	6.0830E+00	2.5412E-01	8.2275E-04	5.7297E+00
F25	ave	2.7040E+03	2.7021E+03	2.7079E+03	2.7016E+03	2.7059E+03	2.7006E+03	2.7000E+03	**2.7000E+03**	2.7072E+03	2.7000E+03	**2.7000E+03**	2.7028E+03
	std	4.5973E+00	3.4123E+00	1.5102E+00	3.7060E+00	1.5918E+00	1.9210E+00	2.0035E-02	**0.0000E+00**	2.2152E+00	1.5514E-02	**0.0000E+00**	5.3742E-02
F26	ave	2.7270E+03	2.7203E+03	2.7036E+03	2.7535E+03	2.7003E+03	2.7138E+03	2.7003E+03	2.7240E+03	2.7004E+03	2.7039E+03	2.7436E+03	**2.7002E+03**
	std	4.4762E+01	4.0537E+01	1.8260E+01	5.0659E+01	6.9541E-02	3.4413E+01	7.0977E-02	4.2681E+01	1.0457E-01	1.8151E+01	5.0179E+01	**1.3289E-02**
F27	ave	3.1549E+03	3.2273E+03	3.0726E+03	3.1860E+03	3.1146E+03	3.3109E+03	**3.0127E+03**	3.3004E+03	3.2795E+03	3.1687E+03	3.3038E+03	3.0985E+03
	std	9.2434E+01	1.1439E+02	4.9696E+01	9.4084E+01	5.5302E+01	1.5552E+02	1.9350E+02	1.1111E+02	1.2049E+02	3.8272E+02	1.5187E+02	**9.6613E+00**
F28	ave	3.8322E+03	3.7026E+03	3.6496E+03	3.7994E+03	3.7799E+03	3.9758E+03	**3.0036E+03**	3.9437E+03	3.7083E+03	3.4012E+03	3.7578E+03	3.6262E+03
	std	2.0017E+02	8.2774E+01	4.6483E+01	5.5129E+01	4.8478E+01	2.4884E+02	**9.5882E+00**	1.9960E+02	4.3937E+01	8.8065E+02	9.5056E+01	3.5978E+01
F29	ave	2.8487E+05	2.8475E+05	4.6836E+03	4.4450E+03	4.1009E+03	3.2131E+06	8.8063E+06	4.5828E+03	2.8501E+05	4.9121E+06	8.7855E+05	**3.4675E+03**
	std	1.5337E+06	1.5337E+06	4.6362E+02	3.9739E+02	3.8478E+02	4.6400E+06	2.9315E+06	4.9047E+02	1.5337E+06	6.8964E+06	2.6680E+06	**1.2119E+02**
F30	ave	6.5309E+03	5.6719E+03	6.6103E+03	6.4641E+03	6.1705E+03	6.8146E+03	1.6607E+04	9.5238E+03	7.9736E+03	8.1463E+04	5.9954E+03	**4.2549E+03**
	std	1.2451E+03	1.0686E+03	1.2007E+03	1.0999E+03	1.7112E+03	2.6400E+03	2.5656E+04	7.0469E+03	5.5652E+03	6.4202E+04	1.0583E+03	**1.8012E+02**

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Table 5. Performance Comparison of the Proposed Algorithms on the 50-Dimensional CEC2014 Benchmark.

ID	Metric	SAO	SBOA	ESC	GRO	CBSO	TLBO	ALA	SOO	PGA	SGA	SNS	MESNS
F1	ave	8.7079E+06	8.0859E+06	1.6753E+07	2.6821E+07	1.1132E+07	3.3021E+06	2.3528E+07	1.9723E+07	1.1560E+07	1.1812E+08	1.1569E+07	9.2966E+05
	std	3.2274E+06	3.4259E+06	7.2967E+06	1.2785E+07	5.5863E+06	1.1078E+06	1.2595E+07	1.3083E+07	4.6026E+06	3.9674E+07	3.5095E+06	1.6640E+05
F2	ave	7.4540E+03	2.4347E+04	**6.0934E+03**	7.1271E+08	3.5525E+08	7.9621E+03	1.4865E+06	1.1229E+09	1.0122E+04	4.9082E+09	9.3413E+05	**2.5848E+02**
	std	7.7377E+03	2.8080E+04	**6.0616E+03**	8.5043E+08	1.5162E+08	7.1208E+03	1.7994E+06	9.6657E+08	9.5482E+03	1.7791E+09	4.3323E+05	**4.8306E+01**
F3	ave	3.4201E+04	2.1634E+04	4.5783E+04	6.6886E+04	1.1498E+04	4.0778E+04	2.3763E+04	2.9476E+04	3.0711E+04	1.0507E+05	1.3093E+04	**3.3192E+02**
	std	1.4744E+04	6.3079E+03	8.0994E+03	1.2381E+04	4.8032E+03	7.9038E+03	5.2563E+03	7.1258E+03	1.2709E+04	1.7788E+04	3.7617E+03	**8.4812E+00**
F4	ave	5.0563E+02	5.6533E+02	5.4653E+02	7.7178E+02	6.5436E+02	5.5732E+02	5.3595E+02	8.9980E+02	5.2293E+02	1.4396E+03	6.1414E+02	**4.7360E+02**
	std	2.2894E+01	4.0454E+01	2.4551E+01	1.0954E+02	6.1002E+01	4.7074E+01	3.4878E+01	1.5450E+02	3.6500E+01	2.5907E+02	4.6745E+01	**2.1025E+01**
F5	ave	5.2119E+02	5.2027E+02	5.2118E+02	5.2045E+02	5.2067E+02	5.2118E+02	5.2097E+02	5.2118E+02	**5.2011E+02**	5.2064E+02	5.2077E+02	5.2021E+02
	std	3.7436E-02	9.8954E-02	4.1404E-02	8.2382E-02	1.3064E-01	4.5636E-02	1.9368E-01	3.0307E-02	5.9832E-02	1.6721E-01	**2.8607E-02**	3.7719E-02
F6	ave	6.1267E+02	6.2587E+02	**6.1192E+02**	6.3312E+02	6.4621E+02	6.3668E+02	6.4253E+02	6.2476E+02	6.2719E+02	6.6563E+02	6.2946E+02	6.1398E+02
	std	4.1553E+00	3.3211E+00	3.2713E+00	3.2876E+00	4.0947E+00	4.0173E+00	7.2003E+00	3.7074E+00	5.1355E+00	5.0828E+00	3.4095E+00	**1.1669E+00**
F7	ave	7.0001E+02	7.0009E+02	7.0003E+02	7.0533E+02	7.0386E+02	7.0016E+02	7.0068E+02	7.1344E+02	7.0001E+02	7.6912E+02	7.0069E+02	**7.0000E+02**
	std	8.6529E-03	7.9260E-02	3.8881E-02	2.2797E+00	1.1291E+00	4.2863E-01	1.7580E-01	8.5567E+00	9.8806E-03	3.9658E+01	1.5587E-01	**2.4568E-04**
F8	ave	8.9674E+02	9.3242E+02	**8.3845E+02**	9.8303E+02	9.6397E+02	9.6006E+02	9.3954E+02	9.0896E+02	9.2213E+02	1.1492E+03	9.4096E+02	8.6697E+02
	std	1.9838E+01	2.8382E+01	7.7261E+00	2.0635E+01	2.9887E+01	2.5502E+01	2.9356E+01	2.0087E+01	2.8247E+01	3.6145E+01	1.9594E+01	**7.5283E+00**
F9	ave	1.0277E+03	1.0608E+03	1.0804E+03	1.1020E+03	1.1055E+03	1.0760E+03	1.0896E+03	1.0233E+03	1.0617E+03	1.3525E+03	1.1141E+03	**9.6865E+02**
	std	5.3179E+01	3.0376E+01	4.6319E+01	3.8345E+01	3.0859E+01	3.4016E+01	3.4790E+01	2.3940E+01	3.2013E+01	6.3492E+01	6.0844E+01	**4.9721E+00**
F10	ave	5.4835E+03	3.4037E+03	**1.8957E+03**	6.2196E+03	6.0308E+03	5.7144E+03	7.0810E+03	3.4328E+03	5.8601E+03	9.6851E+03	4.9710E+03	5.1785E+03
	std	8.1174E+02	5.8053E+02	3.9059E+02	8.4629E+02	7.8673E+02	1.4676E+03	1.0965E+03	5.5289E+02	9.9606E+02	1.1882E+03	3.3587E+02	**2.5573E+02**
F11	ave	6.5812E+03	6.7705E+03	1.2097E+04	7.5980E+03	8.2407E+03	1.4540E+04	9.7890E+03	1.3952E+04	7.5358E+03	1.0843E+04	1.0387E+04	**6.2039E+03**
	std	1.0406E+03	7.9789E+02	7.5562E+02	8.4215E+02	6.8498E+02	4.3535E+02	1.1372E+03	4.8348E+02	9.8748E+02	1.2036E+03	**3.4432E+02**	3.5523E+02
F12	ave	1.2031E+03	**1.2003E+03**	1.2032E+03	1.2009E+03	1.2007E+03	1.2038E+03	1.2018E+03	1.2036E+03	1.2004E+03	1.2024E+03	1.2014E+03	1.2006E+03
	std	1.3949E+00	1.6232E-01	3.9327E-01	3.4229E-01	2.1354E-01	2.6956E-01	6.8549E-01	4.2289E-01	2.4058E-01	7.8578E-01	1.6315E-01	**8.6596E-02**
F13	ave	1.3007E+03	1.3006E+03	1.3004E+03	1.3005E+03	1.3005E+03	1.3006E+03	1.3004E+03	1.3006E+03	1.3006E+03	1.3006E+03	1.3006E+03	**1.3003E+03**
	std	8.8708E-02	1.0194E-01	6.9088E-02	1.0023E-01	6.6787E-02	8.3135E-02	8.5394E-02	8.3088E-02	9.8549E-02	1.2238E-01	8.3224E-02	**2.9984E-02**
F14	ave	1.4006E+03	1.4004E+03	1.4004E+03	1.4011E+03	1.4003E+03	1.4003E+03	1.4004E+03	1.4008E+03	1.4006E+03	1.4047E+03	1.4003E+03	**1.4002E+03**
	std	2.7507E-01	1.5271E-01	1.1076E-01	4.1963E+00	2.0846E-02	1.0499E-01	1.8942E-01	2.2816E+00	2.8165E-01	7.0409E+00	3.4956E-02	**2.0385E-02**
F15	ave	1.5374E+03	1.5263E+03	1.5241E+03	1.8849E+03	1.5486E+03	1.5623E+03	1.5338E+03	2.1860E+03	1.5182E+03	3.1914E+03	1.5372E+03	**1.5078E+03**
	std	3.6917E+00	6.0649E+00	3.1660E+00	3.5330E+02	1.5843E+01	2.3489E+01	9.0431E+00	7.0284E+02	3.4684E+00	1.3617E+03	9.4349E+00	**8.3412E-01**
F16	ave	1.6225E+03	1.6201E+03	1.6218E+03	1.6204E+03	1.6211E+03	1.6220E+03	1.6221E+03	1.6216E+03	1.6219E+03	1.6220E+03	1.6207E+03	**1.6194E+03**
	std	3.0615E-01	7.2146E-01	3.2690E-01	6.1081E-01	5.7201E-01	3.0296E-01	4.0489E-01	3.7058E-01	7.1838E-01	5.5958E-01	**2.0826E-01**	4.7017E-01
F17	ave	8.5674E+05	1.5016E+06	2.5712E+06	1.8865E+06	9.0954E+03	9.3986E+05	2.4748E+05	1.4944E+06	1.6511E+06	1.0046E+07	1.7440E+06	**3.7393E+03**
	std	4.7230E+05	1.0585E+06	1.2871E+06	1.1454E+06	3.6297E+03	7.4415E+05	1.3942E+05	9.0936E+05	8.4270E+05	5.0261E+06	7.9835E+05	**1.9565E+02**
F18	ave	3.3174E+03	4.7309E+03	3.2383E+03	3.4433E+03	2.2351E+03	4.1780E+03	4.6065E+03	3.4887E+03	3.9258E+03	1.2736E+06	4.1419E+03	**2.0252E+03**
	std	1.5993E+03	1.8566E+03	9.9645E+02	1.0760E+03	8.0332E+01	1.4890E+03	2.3263E+03	1.3061E+03	1.9936E+03	1.9350E+06	1.7114E+03	**4.5271E+01**

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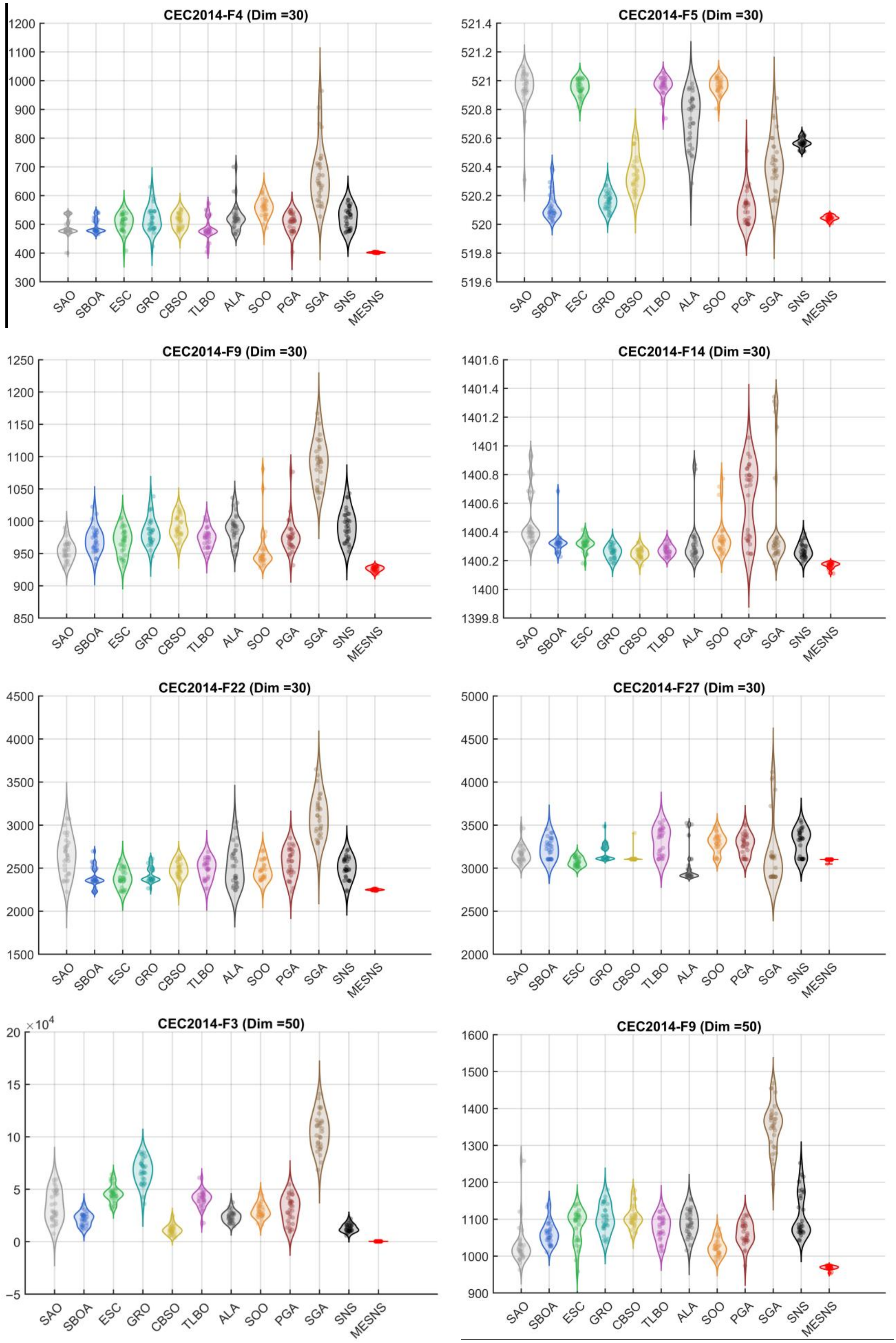
F19	ave	1.9329E+03	1.9551E+03	1.9571E+03	1.9551E+03	1.9498E+03	1.9443E+03	1.9608E+03	1.9746E+03	1.9396E+03	2.0007E+03	1.9576E+03	**1.9160E+03**
	std	1.5508E+01	2.4689E+01	1.4550E+01	2.6050E+01	2.1231E+01	2.8938E+01	2.2616E+01	2.1740E+01	1.8465E+01	3.7690E+01	3.1056E+01	**1.4323E+00**
F20	ave	3.6925E+04	9.8237E+03	2.1498E+04	2.7018E+04	2.4633E+03	1.3166E+04	3.0807E+03	9.8648E+03	3.0980E+04	4.6414E+04	1.6066E+04	**2.1325E+03**
	std	1.4472E+04	3.9893E+03	7.0327E+03	8.0184E+03	1.2233E+02	4.7868E+03	5.5504E+02	4.5501E+03	1.1578E+04	1.4676E+04	4.3753E+03	**1.4468E+01**
F21	ave	3.2740E+05	9.0039E+05	2.7920E+06	9.5128E+05	9.7111E+03	6.7777E+05	5.2961E+04	8.8125E+05	1.1687E+06	7.6624E+06	8.4051E+05	**3.8334E+03**
	std	1.6488E+05	5.1317E+05	1.7030E+06	5.1214E+05	4.2404E+03	5.5948E+05	4.0389E+04	5.0594E+05	6.4291E+05	3.9316E+06	4.9858E+05	**1.0592E+02**
F22	ave	3.2080E+03	3.0521E+03	2.8999E+03	2.9183E+03	3.1662E+03	2.9025E+03	3.2996E+03	2.9601E+03	3.1148E+03	3.8494E+03	3.0086E+03	**2.5428E+03**
	std	2.8404E+02	2.6302E+02	2.5909E+02	2.7764E+02	2.1934E+02	2.8443E+02	2.4279E+02	4.2381E+02	3.3287E+02	4.0055E+02	2.6917E+02	**8.4656E+01**
F23	ave	2.6440E+03	2.6344E+03	2.6461E+03	2.6480E+03	2.6513E+03	2.6440E+03	2.5248E+03	2.6030E+03	2.6440E+03	**2.5237E+03**	2.5288E+03	2.6440E+03
	std	6.0609E-04	3.6535E+01	3.9002E+00	2.7206E+00	3.5953E+00	4.6311E-03	2.5843E+01	7.4180E+01	1.9135E-02	6.7480E+01	5.8632E+01	**3.0207E-08**
F24	ave	2.6732E+03	2.6000E+03	2.6763E+03	2.6022E+03	2.6810E+03	2.6000E+03	2.6002E+03	**2.6000E+03**	2.6803E+03	2.6005E+03	2.6000E+03	2.6656E+03
	std	1.8686E+01	4.6252E-04	3.1730E+00	1.2195E+01	4.2933E+00	3.2555E-03	1.6949E-01	**6.3068E-05**	4.8832E+00	3.9853E-01	8.7757E-04	6.4741E+00
F25	ave	2.7061E+03	**2.7000E+03**	2.7220E+03	2.7019E+03	2.7196E+03	2.7000E+03	2.7000E+03	**2.7000E+03**	2.7157E+03	2.7000E+03	**2.7000E+03**	2.7052E+03
	std	9.2493E+00	**0.0000E+00**	4.2235E+00	7.4094E+00	6.2795E+00	4.2222E-13	3.3466E-03	**0.0000E+00**	3.1247E+00	4.1965E-02	**0.0000E+00**	3.2241E+00
F26	ave	2.7770E+03	2.7801E+03	2.7671E+03	2.7969E+03	2.7003E+03	2.7470E+03	2.7004E+03	2.7671E+03	2.7006E+03	2.7503E+03	2.7834E+03	**2.7003E+03**
	std	4.2808E+01	4.0500E+01	7.3097E+01	1.8240E+01	5.4748E-02	5.0515E+01	1.0126E-01	4.7361E+01	1.0685E-01	5.0510E+01	3.7697E+01	**2.3198E-02**
F27	ave	3.4338E+03	3.7738E+03	3.3350E+03	3.8252E+03	4.0678E+03	4.0919E+03	**3.1685E+03**	3.7814E+03	3.6783E+03	3.6111E+03	3.8362E+03	3.4627E+03
	std	1.2441E+02	1.2028E+02	8.6625E+01	8.4889E+01	2.9380E+02	8.6005E+01	4.7361E+02	8.8347E+01	1.1752E+02	9.5039E+02	1.0108E+02	**3.3854E+01**
F28	ave	4.3758E+03	4.1280E+03	4.2092E+03	5.0228E+03	4.3688E+03	5.1181E+03	**3.1360E+03**	5.3632E+03	4.1106E+03	4.7479E+03	4.5124E+03	4.0903E+03
	std	4.7084E+02	1.5239E+02	1.5471E+02	5.5807E+02	1.7966E+02	7.0141E+02	2.6105E+02	4.7063E+02	1.2477E+02	2.3385E+03	3.6646E+02	**5.3702E+01**
F29	ave	1.0380E+07	5.1748E+03	1.0431E+04	1.1292E+04	1.9017E+04	4.0366E+07	5.3945E+07	3.3006E+04	4.9297E+06	1.6854E+07	1.7005E+07	**4.4032E+03**
	std	1.7556E+07	6.5984E+02	5.1806E+03	7.1647E+03	9.9976E+03	3.1223E+07	1.1346E+07	6.6092E+04	1.2792E+07	2.9407E+07	1.8499E+07	**2.1202E+02**
F30	ave	1.5891E+04	1.4023E+04	2.1784E+04	2.5459E+04	2.4759E+04	1.9370E+04	8.9759E+04	4.5742E+04	1.7071E+04	9.0639E+05	1.7714E+04	**1.1752E+04**
	std	2.4779E+03	1.5848E+03	4.7334E+03	4.3035E+03	6.1146E+03	1.5227E+04	2.2134E+05	2.5652E+04	2.6948E+03	2.4221E+06	2.8672E+03	**2.4567E+02**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CEC2014-F5 (Dim =30)
CEC2014-F9 (Dim =30)
CEC2014-F14 (Dim =30)
CEC2014-F22 (Dim =30)
CEC2014-F27 (Dim =30)
CEC2014-F3 (Dim =50)
CEC2014-F9 (Dim =50)
SAO SBOA ESC GRO CBSO TLBO ALA SOO PGA SGA SNS MESNS

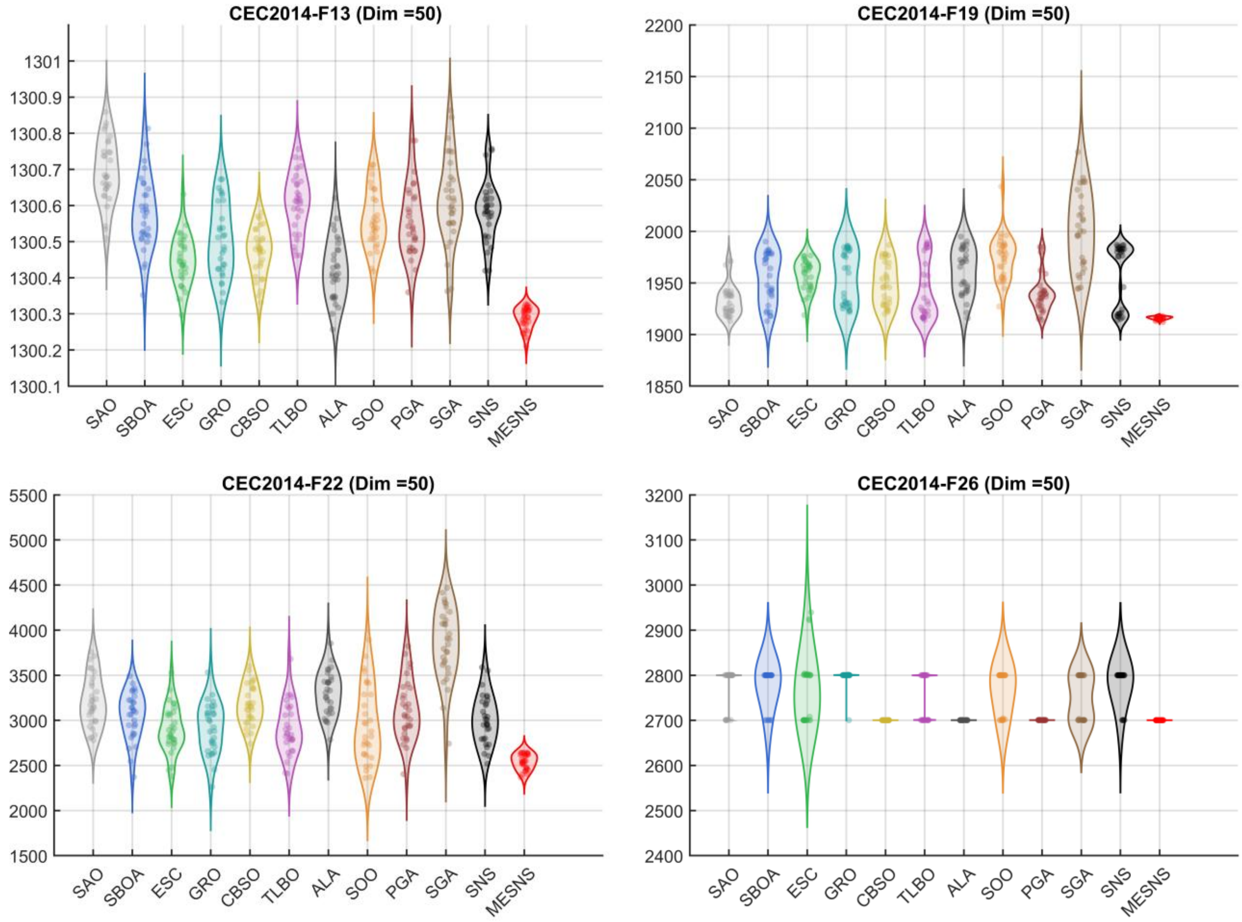


Figure 3. Performance Distribution of Different Algorithms on the CEC2014 Test Functions.

4.3. Numerical Results and Analysis on the CEC2017 Benchmark

To verify the performance of the proposed MESNS, the IEEE CEC2017 benchmark suite was used as the evaluation platform in this section. Two dimensional scenarios were designed, namely 30-dimensional and 50-dimensional, and comparative experiments were conducted with 11 other classic metaheuristic algorithms. To systematically evaluate the global exploration and local exploitation capabilities of MESNS, the focus was on three core performance indicators: optimization accuracy, convergence speed, and result stability. The experimental results are shown below. Tables 6 and 7 summarize the mean and standard deviation of each algorithm after 30 runs, where ave represents the mean of 30 runs and std represents the standard deviation of 30 runs. Figure 4 shows the convergence curves of each algorithm on the test set, and Figure 5 shows the box plots of the results of 30 runs of each algorithm. The optimal experimental results for each function are shown in bold.

As shown in Tables 6 and 7, the CEC2017 benchmark function experimental results demonstrate that the MESNS algorithm exhibits excellent optimization performance and good stability at both 30-dimensional and 50-dimensional scales. In the 30-dimensional case, MESNS achieves lower mean values (ave) and smaller standard deviations (std) on most unimodal, multimodal, and mixed/combined functions, indicating that it not only obtains highly accurate optimal solutions but also possesses strong operational stability and robustness. Particularly on complex multimodal and combined functions, MESNS shows a more significant advantage over other comparative algorithms, demonstrating a more effective global exploration and local development coordination mechanism in complex nonlinear search spaces. When the dimension increases to 50, the overall performance of all algorithms decreases, but MESNS still maintains leading or competitive results on most test functions, with a relatively small increase in mean value and a low standard deviation, demonstrating good high-dimensional adaptability and scalability. Comprehensive analysis shows that MESNS' performance on the CEC2017 test set is consistent with the experimental results of CEC2014, further verifying the stability, effectiveness, and comprehensive optimization capabilities of the algorithm under different test platforms and different dimensional scales.

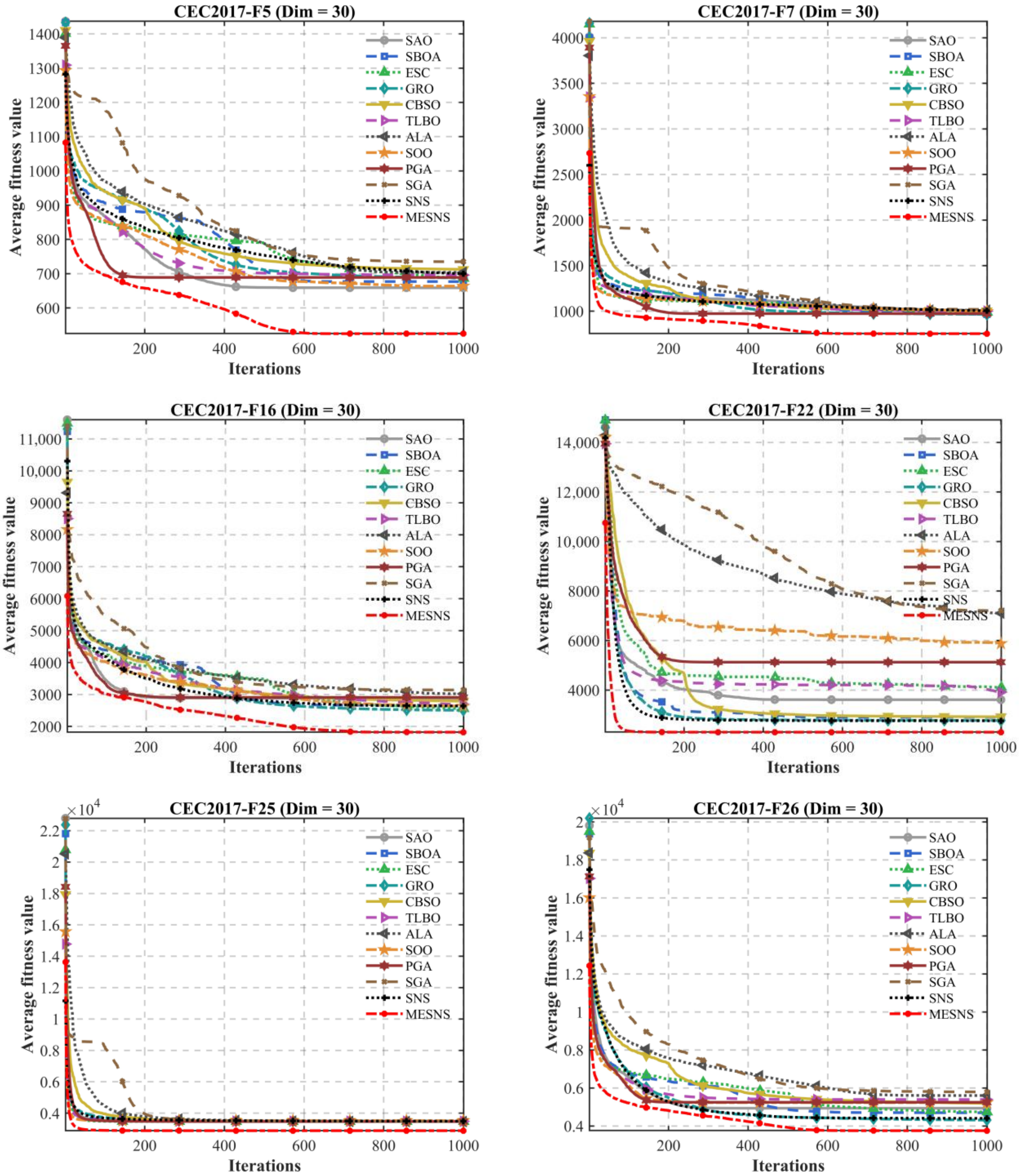
Figure 4 shows the convergence curves, which visually reflect the dynamic optimization process of different algorithms on the CEC2017 test function. Overall, MESNS exhibits faster convergence speed and better final convergence accuracy on most functions. In the early stages of iteration, the MESNS curve drops rapidly, significantly reducing the objective function value in fewer iterations, demonstrating strong global search capability and high search efficiency. In the mid-to-late stages, its convergence curve remains stable and continues to decline without significant oscillations or prolonged stagnation, indicating that the algorithm possesses good local exploitation and fine-grained search capabilities in approaching the optimal solution. In contrast, some of the compared algorithms, although showing a certain downward trend in the early stages, subsequently experienced a significant slowdown in convergence speed, and even exhibited premature convergence on complex multi-peak or combined functions, making it difficult to further improve the quality of the solution. Furthermore, in high-dimensional cases, the convergence process of all algorithms generally slows down, but MESNS still maintains a relatively stable downward trend and a better final value level, indicating that it has good adaptability and scalability in complex high-dimensional search spaces. In summary, the convergence characteristic analysis in Figure 4 further verifies the comprehensive advantages of MESNS in terms of search efficiency, convergence speed, and solution quality, which is consistent with the aforementioned statistical experimental results.

The performance distribution results shown in Figure 5 provide a more comprehensive evaluation of the overall performance of different algorithms on the CEC2017 test functions. As can be seen from the figure, MESNS generally performs well on most test functions, with a relatively lower and more concentrated distribution range of its objective function values, indicating that the algorithm can consistently obtain high-quality solutions on different types of functions. Compared with other comparative algorithms, MESNS exhibits less dispersion and fewer extreme values, indicating lower fluctuations and stronger robustness. However, some algorithms show a wider performance distribution range, exhibiting a significant long-tail phenomenon on complex multimodal and combined functions, reflecting their susceptibility to performance degradation or instability on certain problems. Furthermore, in high-dimensional test scenarios, MESNS maintains a relatively compact and prominent distribution, further demonstrating its adaptability and stability in complex high-dimensional search spaces. Overall, the performance distribution characteristics presented in Figure 5 statistically validate the comprehensive advantages of MESNS in terms of solution quality, consistency, and robustness, consistent with the aforementioned analysis results of the mean, standard deviation, and convergence curve.

In addition to the benchmark function evaluations, it is important to analyze the convergence behavior of optimization algorithms in real-world large-scale combinatorial problems. For instance, the space surveillance network (SSN) scheduling problem is a typical NP-hard large-scale optimization task, characterized by high-dimensional decision variables and complex constraints.

An improved particle swarm optimization algorithm, namely RNP-PSO [29], has been proposed to address this problem by incorporating reverse learning, neighbor adjustment, and particle perturbation strategies. Experimental results on the SSN scheduling problem demonstrate that the improved PSO significantly enhances convergence performance and solution quality. Specifically, the task completion rate (TCR) is improved by approximately 17–19%, while the resource consumption rate (RCR) is reduced by over 30–50%, indicating strong optimization capability in large-scale scheduling scenarios.

These findings suggest that advanced metaheuristic algorithms with enhanced exploration–exploitation balance can achieve superior convergence performance not only on benchmark functions but also in complex real-world optimization problems. Therefore, the convergence advantages observed for the proposed MESNS algorithm on benchmark functions are expected to generalize to large-scale scheduling and deployment problems.



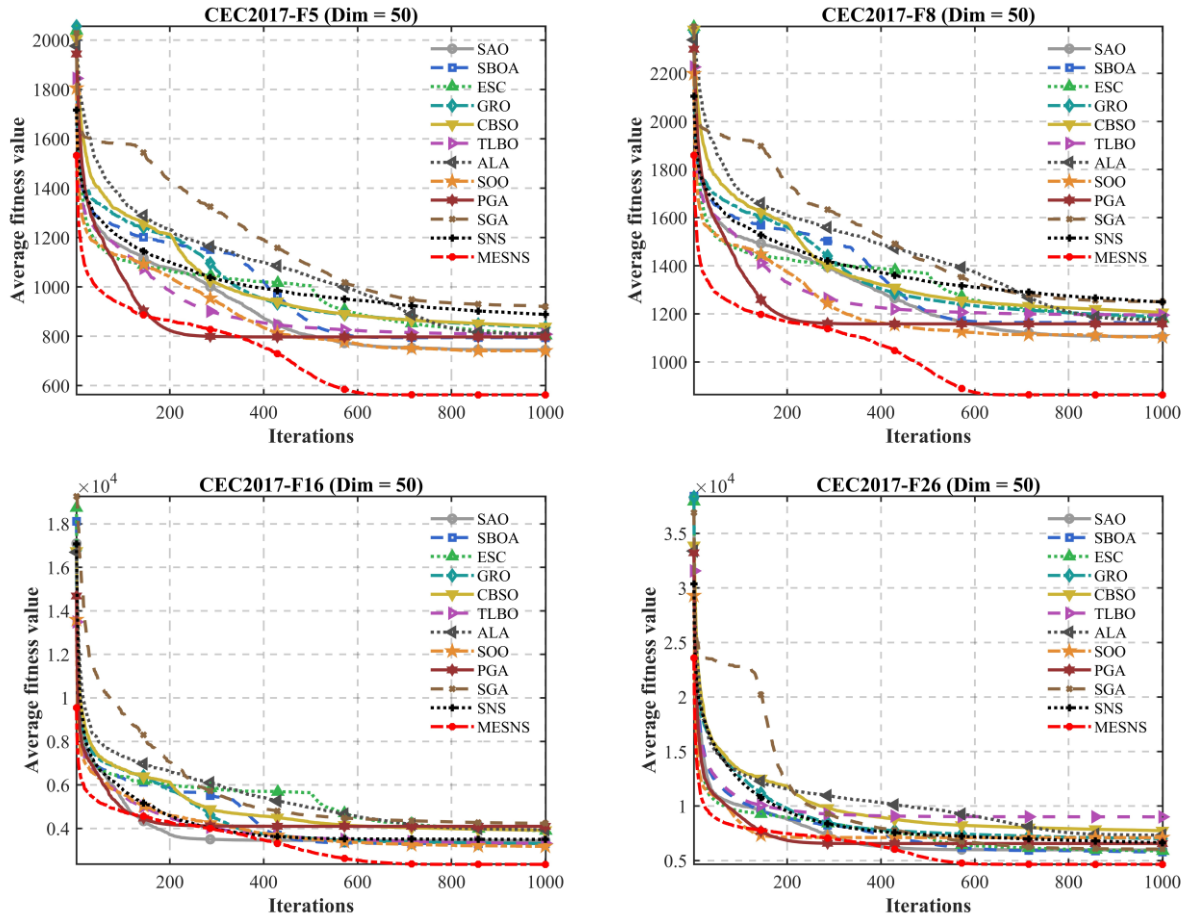
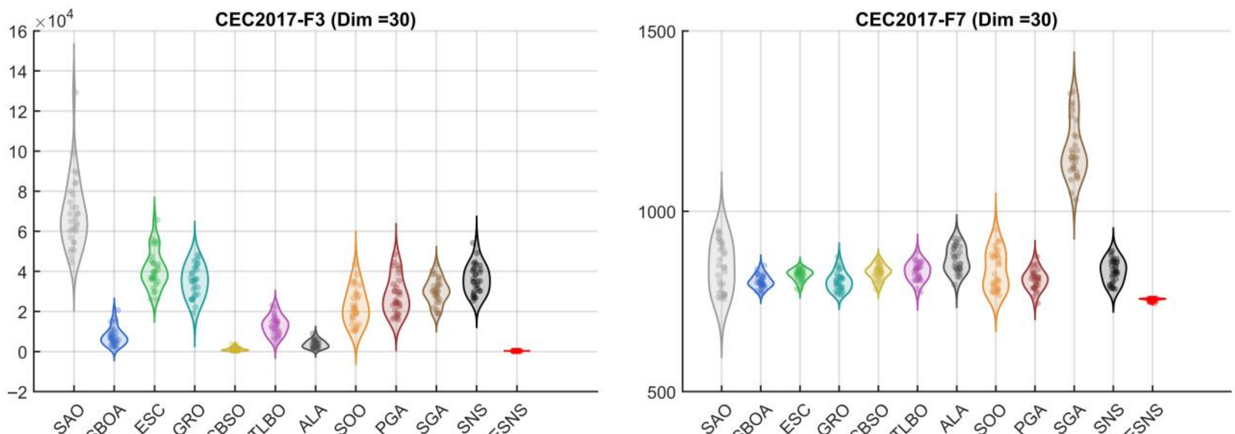


Figure 4. Convergence Speed Comparison of Different Algorithms on the CEC2017 Test Functions.



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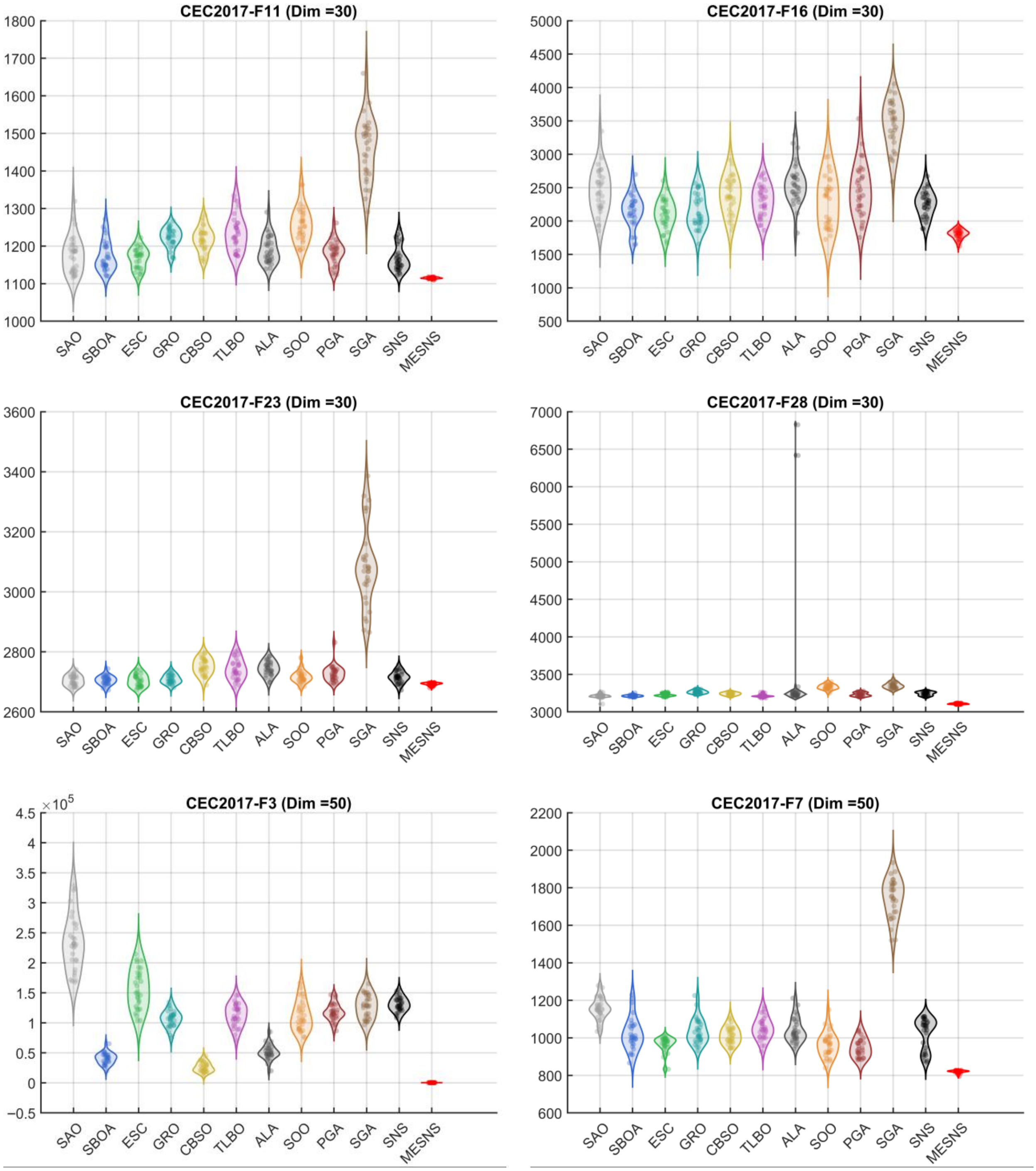
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CEC2017-F28 (Dim =30)
CEC2017-F3 (Dim =50)
CEC2017-F7 (Dim =50)
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SBOA
ESC
GRO
CBSO
TLBO
ALA
SOO
PGA
SGA
SNS
MESNS

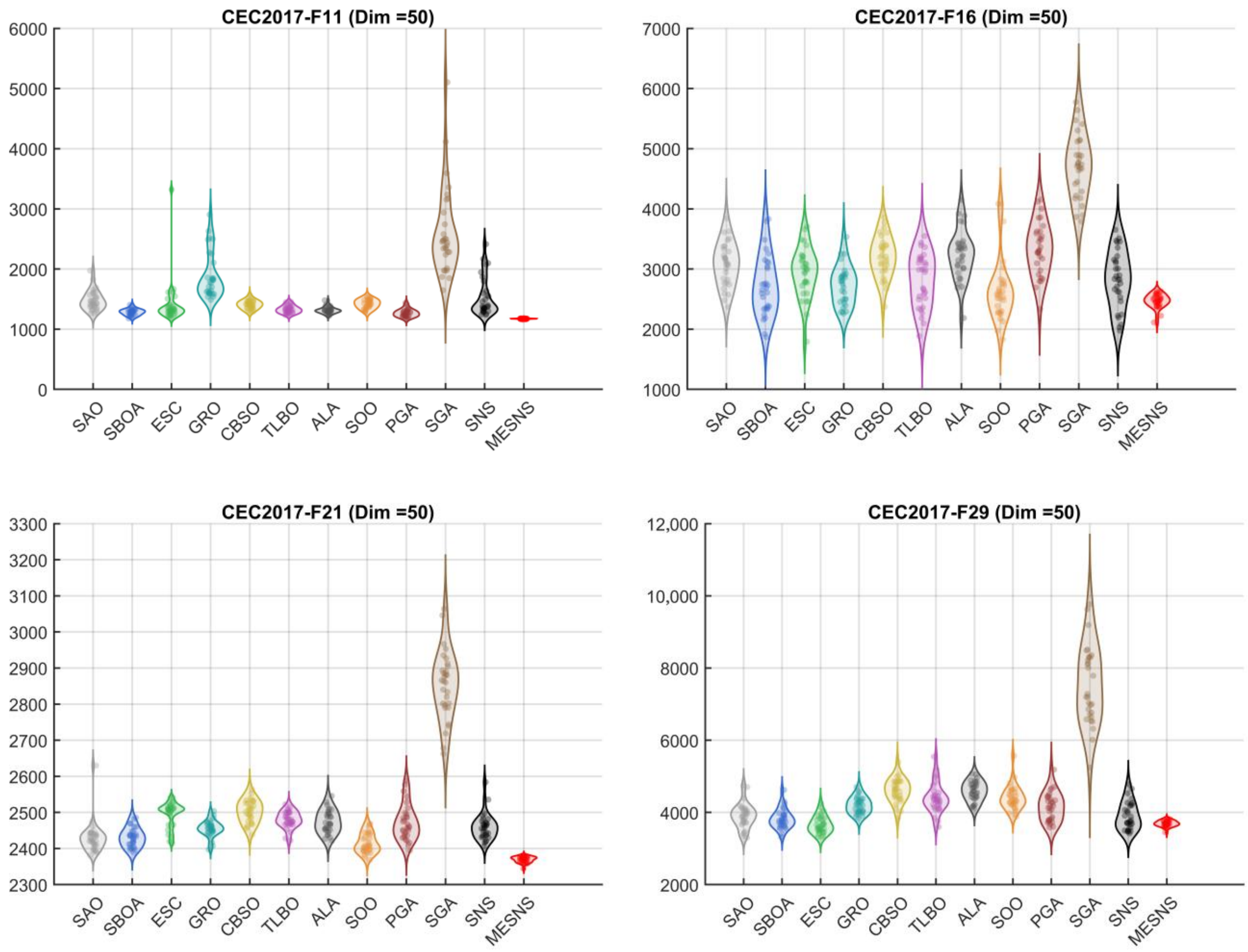

Figure 5. Performance Distribution of Different Algorithms on the CEC2017 Test Functions.

4.4. Numerical Results and Analysis on the CEC2022 Benchmark

To verify the performance of the proposed MESNS, the IEEE CEC2022 benchmark suite was used as the evaluation platform in this section. One dimensional scenario was designed, namely 10-dimensional, and comparative experiments were conducted with 11 other classic metaheuristic algorithms. To systematically evaluate the global exploration and local exploitation capabilities of MESNS, the focus was on three core performance indicators: optimization accuracy, convergence speed, and result stability. The experimental results are shown below. Table 8 summarize the mean and standard deviation of each algorithm after 30 runs, where ave represents the mean of 30 runs and std represents the standard deviation of 30 runs. Figure 6 shows the convergence curves of each algorithm on the test set, and Figure 7 shows the box plots of the results of 30 runs of each algorithm. The optimal experimental results for each function are shown in bold.

Table 6. Performance Comparison of the Proposed Algorithms on the 30-Dimensional CEC2017 Benchmark.

ID	Metric	SAO	SBOA	ESC	GRO	CBSO	TLBO	ALA	SOO	PGA	SGA	SNS	MESNS
F1	ave	5.1954E+03	4.5276E+03	3.9260E+03	1.5656E+06	1.0562E+06	3.2920E+03	4.0619E+03	2.4102E+06	7.4666E+03	3.8439E+08	4.5657E+03	**1.0008E+02**
	std	5.4325E+03	4.9474E+03	3.7451E+03	1.2003E+06	9.0225E+05	3.6751E+03	4.1435E+03	5.1182E+06	6.5843E+03	3.6035E+08	3.2182E+03	**8.7972E-02**
F2	ave	9.7906E+15	**7.8181E+12**	2.2027E+14	1.1298E+23	2.6238E+17	7.1117E+14	1.1472E+18	1.6576E+21	1.6028E+13	6.3906E+26	1.7204E+12	2.3041E+05
	std	5.2163E+16	2.9171E+13	9.5258E+14	5.5315E+23	8.8115E+17	3.7914E+15	4.4031E+18	6.4414E+21	**2.8477E+13**	2.4406E+27	4.7745E+12	2.3549E+05
F3	ave	6.8918E+04	7.5872E+03	4.1173E+04	3.5062E+04	1.2232E+03	1.3045E+04	3.7120E+03	2.2610E+04	2.8061E+04	3.0031E+04	3.6183E+04	**3.0002E+02**
	std	1.7757E+04	4.5440E+03	9.5767E+03	7.8769E+03	8.5459E+02	4.8151E+03	1.9618E+03	8.9579E+03	9.3781E+03	6.4700E+03	7.1758E+03	**4.8515E-03**
F4	ave	4.9006E+02	4.8962E+02	5.0678E+02	5.1286E+02	4.9979E+02	4.7502E+02	4.9302E+02	5.4729E+02	4.9813E+02	5.7543E+02	5.0368E+02	**4.1725E+02**
	std	1.8303E+01	2.9497E+01	1.3443E+01	1.4742E+01	1.5196E+01	3.2153E+01	**1.2513E+01**	2.3407E+01	1.5166E+01	4.7628E+01	2.3458E+01	1.4657E+01
F5	ave	5.5739E+02	5.5859E+02	5.8315E+02	5.7416E+02	5.9323E+02	5.8312E+02	5.9493E+02	5.4935E+02	5.7518E+02	7.4888E+02	5.8380E+02	**5.2797E+02**
	std	1.8025E+01	2.0619E+01	1.5255E+01	1.4739E+01	2.2701E+01	1.7695E+01	2.8370E+01	2.1012E+01	2.1669E+01	5.4511E+01	2.8835E+01	**3.8818E+00**
F6	ave	6.0001E+02	6.0081E+02	6.0000E+02	6.0389E+02	6.1097E+02	6.0796E+02	6.0399E+02	6.0221E+02	6.0003E+02	6.6432E+02	6.0006E+02	**6.0000E+02**
	std	3.2187E-02	1.9408E+00	3.5102E-03	1.5348E+00	3.9695E+00	3.8614E+00	2.2812E+00	9.7306E-01	3.5660E-02	7.6994E+00	3.6036E-02	**3.9994E-04**
F7	ave	8.4466E+02	8.0522E+02	8.2336E+02	8.0563E+02	8.3079E+02	8.3346E+02	8.6527E+02	8.3674E+02	8.1020E+02	1.1688E+03	8.3385E+02	**7.5458E+02**
	std	6.5982E+01	1.8456E+01	1.5670E+01	2.6421E+01	2.1225E+01	2.5610E+01	3.5285E+01	5.3704E+01	2.8547E+01	8.0834E+01	3.1737E+01	**3.7875E+00**
F8	ave	8.5612E+02	8.6262E+02	8.7217E+02	8.7256E+02	8.8029E+02	8.6423E+02	8.8382E+02	8.4406E+02	8.8938E+02	9.8341E+02	8.8104E+02	**8.2465E+02**
	std	1.5663E+01	1.6118E+01	2.1530E+01	1.9139E+01	2.1392E+01	1.2548E+01	2.1190E+01	2.8022E+01	2.9835E+01	3.4027E+01	1.8272E+01	**3.5632E+00**
F9	ave	9.0543E+02	9.9060E+02	9.0102E+02	1.1822E+03	1.4542E+03	1.1533E+03	1.3838E+03	1.1009E+03	9.1137E+02	5.8153E+03	9.5779E+02	**9.0024E+02**
	std	6.2050E+00	1.6174E+02	1.8897E+00	2.0893E+02	3.3099E+02	3.2663E+02	4.0706E+02	1.8134E+02	1.7648E+01	1.2052E+03	4.6509E+01	**1.7841E-01**
F10	ave	3.8779E+03	4.1123E+03	6.4045E+03	4.2811E+03	4.6422E+03	7.9413E+03	5.0553E+03	7.4192E+03	4.4237E+03	5.7976E+03	5.5663E+03	**3.4111E+03**
	std	7.2155E+02	8.7720E+02	5.0202E+02	6.4849E+02	6.8677E+02	6.0998E+02	6.3409E+02	5.9315E+02	5.8849E+02	7.8889E+02	3.5887E+02	**2.0773E+02**
F11	ave	1.1769E+03	1.1730E+03	1.1670E+03	1.2225E+03	1.2182E+03	1.2343E+03	1.1898E+03	1.2568E+03	1.1813E+03	1.4588E+03	1.1677E+03	**1.1148E+03**
	std	4.9285E+01	4.0449E+01	2.7822E+01	2.6444E+01	3.2986E+01	4.7078E+01	3.5537E+01	4.5567E+01	3.2234E+01	8.2161E+01	3.4641E+01	**2.0435E+00**
F12	ave	2.8929E+05	4.4466E+05	7.9478E+05	6.7892E+05	1.7585E+04	6.3495E+04	4.9866E+05	1.0262E+05	7.7619E+05	3.9006E+07	6.5574E+05	**2.0598E+03**
	std	2.5709E+05	3.0450E+05	5.5743E+05	4.8065E+05	1.2591E+04	5.6044E+04	7.5209E+05	1.5644E+05	6.6719E+05	4.1849E+07	4.3439E+05	**1.1878E+02**
F13	ave	2.0116E+04	1.4130E+04	1.6301E+04	2.2177E+04	2.4575E+03	1.4748E+04	2.2686E+04	2.7221E+04	2.7292E+04	3.2941E+05	1.6337E+04	**1.3724E+03**
	std	1.5518E+04	1.4765E+04	1.1700E+04	1.4901E+04	4.6197E+02	1.4787E+04	2.3205E+04	2.0357E+04	2.0684E+04	6.1963E+05	1.4849E+04	**9.8264E+00**
F14	ave	2.3200E+04	1.1885E+04	5.3306E+04	1.3469E+04	1.4772E+03	1.3773E+04	1.5265E+03	3.7109E+03	6.2111E+04	1.1758E+05	9.1828E+03	**1.4333E+03**
	std	2.4173E+04	1.0498E+04	4.2997E+04	1.2752E+04	1.7066E+01	1.3638E+04	2.5927E+01	6.2252E+03	5.3426E+04	1.9033E+05	8.0371E+03	**2.6305E+00**
F15	ave	4.8731E+03	1.0098E+04	6.4484E+03	6.0690E+03	1.7183E+03	8.6537E+03	5.5760E+03	5.2228E+03	1.6811E+04	9.4332E+04	2.6503E+03	**1.5366E+03**
	std	4.2146E+03	1.0081E+04	6.5469E+03	4.9404E+03	7.1131E+01	8.3842E+03	1.0000E+04	6.3012E+03	1.2711E+04	6.6123E+04	1.2523E+03	**4.7502E+00**
F16	ave	2.4498E+03	2.1582E+03	2.1024E+03	2.1460E+03	2.3846E+03	2.3017E+03	2.5667E+03	2.2847E+03	2.4602E+03	3.4483E+03	2.2546E+03	**1.8037E+03**

	std	3.3499E+02	2.4717E+02	2.2611E+02	2.4654E+02	2.7261E+02	2.3871E+02	3.0647E+02	3.6762E+02	4.0153E+02	3.5396E+02	1.9288E+02	**6.9617E+01**
F17	ave	2.0694E+03	1.8622E+03	1.8194E+03	1.8698E+03	1.9206E+03	1.9114E+03	2.0339E+03	1.9105E+03	2.0647E+03	2.5172E+03	1.8614E+03	**1.7836E+03**
	std	1.9290E+02	7.5589E+01	8.2650E+01	9.0472E+01	8.3500E+01	1.2665E+02	1.5581E+02	1.1605E+02	1.8985E+02	2.5554E+02	1.0279E+02	**1.5695E+01**
F18	ave	2.6613E+05	3.3039E+05	4.2548E+05	2.1163E+05	1.9155E+03	2.6699E+05	4.7925E+03	2.8886E+05	5.7312E+05	3.0674E+06	1.6141E+05	**1.8376E+03**
	std	2.1096E+05	2.8083E+05	3.6617E+05	2.2721E+05	3.9843E+01	1.3692E+05	2.2907E+03	2.9065E+05	3.4210E+05	3.1897E+06	1.1092E+05	**2.2632E+00**
F19	ave	9.1457E+03	1.2181E+04	1.0244E+04	6.1855E+03	1.9598E+03	6.6075E+03	2.7993E+03	3.1864E+03	1.9147E+04	6.3728E+05	7.1575E+03	**1.9176E+03**
	std	1.2522E+04	1.3330E+04	1.0573E+04	5.8825E+03	1.8773E+01	4.5887E+03	2.8072E+03	2.6282E+03	2.0610E+04	6.9228E+05	4.6104E+03	**1.4189E+00**
F20	ave	2.3361E+03	2.1982E+03	2.2034E+03	2.2563E+03	2.3311E+03	2.2832E+03	2.3923E+03	2.2742E+03	2.3027E+03	2.7476E+03	2.1676E+03	**2.0705E+03**
	std	1.6923E+02	1.0920E+02	9.4937E+01	6.1649E+01	1.0613E+02	1.3458E+02	1.5079E+02	1.2992E+02	1.7594E+02	1.5942E+02	9.3129E+01	**1.8574E+01**
F21	ave	2.3569E+03	2.3447E+03	2.3722E+03	2.3578E+03	2.3871E+03	2.3644E+03	2.3931E+03	2.3423E+03	2.3745E+03	2.5536E+03	2.3611E+03	**2.3299E+03**
	std	1.6568E+01	1.3761E+01	1.9216E+01	1.2145E+01	2.5027E+01	2.3735E+01	2.6061E+01	1.0317E+01	2.2481E+01	6.4879E+01	1.7474E+01	**4.2246E+00**
F22	ave	2.6792E+03	2.3005E+03	2.6828E+03	2.3082E+03	2.3312E+03	2.3014E+03	6.0908E+03	2.9857E+03	3.8520E+03	6.8629E+03	2.3010E+03	**2.3000E+03**
	std	9.2057E+02	1.2376E+00	1.4564E+03	4.7902E+00	2.2495E+01	2.1182E+00	1.8081E+03	2.0530E+03	1.8785E+03	2.0717E+03	1.4707E+00	**1.1527E-08**
F23	ave	2.7041E+03	2.7054E+03	2.7053E+03	2.7091E+03	2.7513E+03	2.7412E+03	2.7440E+03	2.7173E+03	2.7276E+03	3.0858E+03	2.7157E+03	**2.6922E+03**
	std	1.7609E+01	1.5105E+01	2.0311E+01	1.4931E+01	2.7336E+01	3.4279E+01	2.3967E+01	2.0653E+01	2.6444E+01	1.3567E+02	1.9455E+01	**4.7905E+00**
F24	ave	2.8745E+03	2.8704E+03	2.9137E+03	2.8728E+03	2.9262E+03	2.8933E+03	2.9249E+03	2.8875E+03	2.8929E+03	3.2654E+03	2.9094E+03	**2.8557E+03**
	std	1.8017E+01	1.5530E+01	1.6218E+01	1.7811E+01	2.5198E+01	1.5161E+01	2.1261E+01	2.3846E+01	1.8492E+01	1.1015E+02	2.8216E+01	**5.4312E+00**
F25	ave	2.8874E+03	2.8942E+03	2.8881E+03	2.9105E+03	2.8991E+03	2.9052E+03	2.8916E+03	2.9335E+03	2.8884E+03	2.9807E+03	2.9106E+03	**2.8835E+03**
	std	1.1953E+00	1.3475E+01	1.7751E+00	1.7551E+01	1.1420E+01	2.3171E+01	1.2174E+01	2.9280E+01	4.7059E+00	2.9520E+01	2.1607E+01	**4.7441E-02**
F26	ave	4.1127E+03	3.9721E+03	4.0353E+03	**3.6055E+03**	4.2009E+03	4.4567E+03	4.6026E+03	4.4227E+03	4.3810E+03	7.1675E+03	3.8092E+03	3.6156E+03
	std	2.9186E+02	5.1208E+02	2.3447E+02	6.7397E+02	7.0114E+02	7.8111E+02	5.7557E+02	2.7448E+02	**2.0698E+02**	2.2008E+03	1.2149E+03	5.3874E+02
F27	ave	3.2201E+03	3.2131E+03	3.2136E+03	3.2360E+03	3.2271E+03	3.2380E+03	3.2234E+03	3.2472E+03	3.2151E+03	3.4208E+03	3.2297E+03	**3.1934E+03**
	std	1.5243E+01	1.0538E+01	8.2653E+00	1.3830E+01	1.1636E+01	2.3698E+01	1.8675E+01	1.9431E+01	1.1542E+01	1.4311E+02	1.4551E+01	**6.6267E+00**
F28	ave	3.2111E+03	3.2132E+03	3.2255E+03	3.2662E+03	3.2402E+03	3.2121E+03	3.4662E+03	3.3316E+03	3.2323E+03	3.3483E+03	3.2437E+03	**3.1074E+03**
	std	2.7390E+01	1.2198E+01	1.8949E+01	2.2077E+01	1.8250E+01	1.8704E+01	8.5991E+02	3.4783E+01	2.4595E+01	3.7707E+01	2.4815E+01	**6.0935E+00**
F29	ave	3.6549E+03	3.4945E+03	3.4879E+03	3.6321E+03	3.7499E+03	3.7087E+03	3.7637E+03	3.6226E+03	3.6838E+03	4.9358E+03	3.4634E+03	**3.4042E+03**
	std	1.6010E+02	1.2278E+02	9.1415E+01	1.3718E+02	1.2869E+02	1.6230E+02	1.7527E+02	1.4327E+02	2.1278E+02	4.0790E+02	9.1451E+01	**3.8415E+01**
F30	ave	8.8485E+03	1.4795E+04	1.0806E+04	1.9056E+04	9.4812E+03	1.1185E+04	2.7424E+04	1.9174E+04	1.0534E+04	5.0097E+06	7.9769E+03	**5.1350E+03**
	std	3.2608E+03	1.4412E+04	3.1159E+03	1.5887E+04	7.3177E+03	3.5972E+03	1.1651E+04	1.4235E+04	4.1880E+03	4.2105E+06	1.6810E+03	**5.7467E+01**

Table 7. Performance Comparison of the Proposed Algorithms on the 50-Dimensional CEC2017 Benchmark.

ID	Metric	SAO	SBOA	ESC	GRO	CBSO	TLBO	ALA	SOO	PGA	SGA	SNS	MESNS
F1	ave	2.7049E+03	1.0458E+04	5.4498E+03	5.1648E+08	2.7985E+08	4.1154E+03	9.4973E+05	7.5009E+08	7.9489E+03	7.1211E+09	1.0966E+06	**1.5520E+02**
	std	3.0296E+03	1.1271E+04	4.4127E+03	3.8705E+08	1.2248E+08	4.5621E+03	9.0378E+05	6.0183E+08	7.4904E+03	3.2576E+09	7.1122E+05	**3.4540E+01**
F2	ave	1.2507E+45	**9.8018E+31**	5.4813E+43	1.2398E+49	8.8341E+42	7.8619E+34	5.7446E+47	4.2028E+46	1.1137E+36	3.1306E+62	5.8163E+39	1.6911E+23
	std	6.8502E+45	**4.7006E+32**	2.9743E+44	6.5393E+49	4.0497E+43	3.0139E+35	3.1464E+48	1.2052E+47	4.7721E+36	1.6969E+63	3.1857E+40	2.0344E+23
F3	ave	2.3395E+05	4.0423E+04	1.5619E+05	1.0635E+05	2.6017E+04	1.1341E+05	5.0989E+04	1.1116E+05	1.1765E+05	1.2733E+05	1.3090E+05	**3.7003E+02**
	std	4.6871E+04	9.3724E+03	3.2463E+04	1.4053E+04	8.0513E+03	1.9015E+04	1.2418E+04	2.4499E+04	1.4324E+04	1.8996E+04	1.1519E+04	**1.5463E+01**
F4	ave	5.4954E+02	5.4118E+02	6.0837E+02	7.1396E+02	6.5009E+02	5.5882E+02	6.2255E+02	8.5887E+02	5.6070E+02	1.2112E+03	5.7182E+02	**4.5433E+02**
	std	3.9242E+01	5.9223E+01	4.2283E+01	7.6603E+01	5.7224E+01	4.3310E+01	4.3491E+01	1.6135E+02	4.7975E+01	2.8119E+02	4.9462E+01	**1.6752E+01**
F5	ave	6.1864E+02	6.6124E+02	6.7179E+02	7.0493E+02	7.0602E+02	6.7903E+02	6.8210E+02	6.1984E+02	6.6494E+02	9.1654E+02	7.3254E+02	**5.6375E+02**
	std	2.9838E+01	3.8104E+01	5.9091E+01	3.3679E+01	3.6479E+01	2.9908E+01	4.3885E+01	2.9013E+01	3.4755E+01	4.3274E+01	7.0785E+01	**6.2641E+00**
F6	ave	6.0047E+02	6.0538E+02	**6.0009E+02**	6.1622E+02	6.2423E+02	6.2079E+02	6.1763E+02	6.1123E+02	6.0287E+02	6.7751E+02	6.0043E+02	6.0011E+02
	std	3.9010E-01	3.5584E+00	9.2884E-02	4.1566E+00	4.9617E+00	4.4821E+00	6.0886E+00	3.9183E+00	3.1451E+00	7.6778E+00	1.3026E-01	**1.9177E-02**
F7	ave	1.1550E+03	1.0222E+03	9.6321E+02	1.0338E+03	1.0126E+03	1.0496E+03	1.0395E+03	9.6338E+02	9.4655E+02	1.7447E+03	1.0225E+03	**8.1944E+02**
	std	5.8307E+01	8.7048E+01	4.4605E+01	6.6753E+01	4.5565E+01	5.9528E+01	6.4969E+01	6.6489E+01	4.9228E+01	1.0467E+02	8.5360E+01	**7.2521E+00**
F8	ave	9.2727E+02	9.5569E+02	9.6058E+02	9.9617E+02	1.0142E+03	9.8759E+02	9.7514E+02	9.1727E+02	9.6829E+02	1.2243E+03	1.0492E+03	**8.6532E+02**
	std	4.1089E+01	3.0236E+01	4.9250E+01	3.2880E+01	3.8432E+01	2.8044E+01	4.0971E+01	2.4494E+01	3.5771E+01	4.4786E+01	5.9543E+01	**6.6660E+00**
F9	ave	9.6897E+02	2.4348E+03	9.5753E+02	3.4735E+03	4.4662E+03	5.9744E+03	3.8978E+03	5.3648E+03	1.3193E+03	1.8804E+04	2.4190E+03	**9.1822E+02**
	std	8.3318E+01	1.0068E+03	7.4689E+01	7.6329E+02	1.2187E+03	2.7032E+03	1.3412E+03	2.5549E+03	3.1836E+02	3.0097E+03	1.8975E+03	**5.8044E+00**
F10	ave	6.6399E+03	6.8481E+03	1.2229E+04	7.1480E+03	8.1341E+03	1.4311E+04	9.8277E+03	1.3813E+04	7.4427E+03	1.0565E+04	9.9914E+03	**6.0792E+03**
	std	1.7422E+03	9.7781E+02	6.6998E+02	9.1509E+02	8.1913E+02	4.5616E+02	1.2415E+03	**3.6736E+02**	8.7523E+02	1.2923E+03	3.8388E+02	4.2072E+02
F11	ave	1.4775E+03	1.2833E+03	1.4145E+03	1.8894E+03	1.4070E+03	1.3288E+03	1.3153E+03	1.4194E+03	1.2698E+03	2.6125E+03	1.5381E+03	**1.1732E+03**
	std	1.6453E+02	5.1888E+01	3.8088E+02	3.8388E+02	7.0894E+01	6.1881E+01	5.4781E+01	7.3739E+01	6.3944E+01	7.3081E+02	3.1984E+02	**5.0342E+00**
F12	ave	3.1266E+06	3.6755E+06	5.8870E+06	1.0301E+07	1.1273E+07	1.1638E+06	7.6283E+06	1.9899E+07	6.9363E+06	5.6743E+08	4.1225E+06	**4.2485E+04**
	std	2.1450E+06	2.6060E+06	3.0988E+06	4.6728E+06	9.2762E+06	8.0236E+05	4.4819E+06	1.7112E+07	3.2019E+06	3.5597E+08	1.6564E+06	**1.2242E+04**
F13	ave	7.4405E+03	1.2371E+04	1.0732E+04	9.3417E+03	1.5802E+04	9.1923E+03	2.5853E+04	2.1157E+04	1.1921E+04	1.0331E+07	9.0078E+03	**2.0147E+03**
	std	7.5042E+03	9.8430E+03	5.8611E+03	4.7365E+03	6.5994E+03	6.1918E+03	1.4416E+04	1.0828E+04	9.6633E+03	1.2524E+07	7.2004E+03	**2.8271E+02**
F14	ave	1.0137E+05	1.2980E+05	4.4852E+05	1.5892E+05	1.6301E+03	9.9497E+04	2.0933E+03	8.6326E+04	1.9376E+05	1.0731E+06	7.3055E+04	**1.4870E+03**
	std	8.0932E+04	1.0100E+05	4.8628E+05	1.6636E+05	2.2447E+01	7.8050E+04	2.5181E+02	6.6841E+04	1.2274E+05	6.8876E+05	6.3863E+04	**8.5686E+00**
F15	ave	1.2987E+04	1.3042E+04	7.3630E+03	8.7121E+03	2.6935E+03	9.1086E+03	1.8984E+04	8.8318E+03	1.3442E+04	2.2433E+05	1.5273E+04	**1.8163E+03**
	std	6.3216E+03	5.0711E+03	5.1351E+03	4.9169E+03	3.1492E+02	6.0687E+03	1.0904E+04	6.7568E+03	1.0029E+04	4.5788E+05	5.0386E+03	**4.2755E+01**
F16	ave	3.0747E+03	2.7229E+03	2.9407E+03	2.7245E+03	3.2019E+03	2.8147E+03	3.2390E+03	2.6522E+03	3.3336E+03	4.7170E+03	2.8293E+03	**2.4704E+03**
	std	3.6297E+02	5.0505E+02	4.2140E+02	3.2427E+02	3.4692E+02	4.4520E+02	4.1827E+02	5.3691E+02	4.4233E+02	5.2185E+02	4.7349E+02	**1.1374E+02**

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F17	ave	2.6492E+03	2.6320E+03	2.6864E+03	2.6870E+03	2.9840E+03	2.7949E+03	3.1074E+03	2.7096E+03	2.8216E+03	3.9233E+03	2.6110E+03	**2.3471E+03**
	std	2.2209E+02	2.8397E+02	2.4080E+02	2.4105E+02	2.7325E+02	3.3169E+02	3.0559E+02	3.7504E+02	3.2387E+02	4.7447E+02	3.4136E+02	**8.0409E+01**
F18	ave	1.4413E+06	1.4551E+06	2.9430E+06	1.8461E+06	3.5991E+03	1.5801E+06	8.2889E+04	1.5067E+06	1.7937E+06	1.4113E+07	1.2320E+06	**1.9938E+03**
	std	1.3296E+06	9.4323E+05	2.5072E+06	1.0779E+06	2.0874E+03	1.1848E+06	3.6443E+04	1.1851E+06	1.4324E+06	1.4310E+07	9.7482E+05	**1.4318E+01**
F19	ave	2.4326E+04	2.1026E+04	1.1949E+04	1.7686E+04	2.1924E+03	1.6368E+04	1.0671E+04	1.7864E+04	1.6231E+04	4.9235E+06	1.9689E+04	**1.9546E+03**
	std	1.2397E+04	1.4265E+04	7.4696E+03	8.1383E+03	2.0618E+02	1.0555E+04	1.0278E+04	1.3063E+04	1.3340E+04	5.8909E+06	9.1305E+03	**7.0751E+00**
F20	ave	2.6774E+03	2.6972E+03	2.8312E+03	2.7130E+03	2.9963E+03	3.3315E+03	3.2098E+03	3.3388E+03	3.0273E+03	3.6402E+03	2.5687E+03	**2.4403E+03**
	std	2.3911E+02	3.3848E+02	2.1972E+02	2.2630E+02	2.2581E+02	3.7003E+02	2.9436E+02	3.2233E+02	3.1080E+02	2.9129E+02	2.1314E+02	**7.7802E+01**
F21	ave	2.4296E+03	2.4308E+03	2.4962E+03	2.4548E+03	2.5024E+03	2.4800E+03	2.4743E+03	2.4141E+03	2.4718E+03	2.8540E+03	2.4595E+03	**2.3693E+03**
	std	4.3876E+01	2.6271E+01	3.4373E+01	2.2810E+01	3.1045E+01	2.3543E+01	3.1855E+01	2.7312E+01	4.9239E+01	9.4150E+01	3.6130E+01	**8.2663E+00**
F22	ave	8.1857E+03	7.4928E+03	1.3316E+04	6.5832E+03	9.7199E+03	1.3154E+04	1.0641E+04	1.5303E+04	9.0969E+03	1.2375E+04	**6.0094E+03**	7.0534E+03
	std	1.6437E+03	2.2454E+03	6.3687E+02	3.3680E+03	1.5139E+03	5.5223E+03	1.3637E+03	**5.4536E+02**	8.4525E+02	1.2068E+03	4.9086E+03	1.6658E+03
F23	ave	2.8464E+03	2.8586E+03	2.8647E+03	2.9298E+03	2.9827E+03	3.0015E+03	2.9300E+03	2.9321E+03	2.8957E+03	3.7956E+03	2.9261E+03	**2.8308E+03**
	std	2.5929E+01	3.3010E+01	5.8969E+01	3.6150E+01	5.1637E+01	5.6607E+01	4.3634E+01	4.1716E+01	2.6254E+01	2.8321E+02	5.2092E+01	**1.3615E+01**
F24	ave	3.0427E+03	3.0233E+03	3.1247E+03	3.0830E+03	3.1367E+03	3.1109E+03	3.1104E+03	3.0935E+03	3.0382E+03	3.9085E+03	3.1088E+03	**2.9999E+03**
	std	8.8193E+01	3.5024E+01	3.5945E+01	3.8980E+01	5.1724E+01	4.0261E+01	5.4324E+01	4.7043E+01	2.6809E+01	1.9295E+02	5.3624E+01	**1.4608E+01**
F25	ave	3.0527E+03	3.1030E+03	3.0987E+03	3.2447E+03	3.1165E+03	3.0795E+03	3.0797E+03	3.3905E+03	3.0562E+03	3.6146E+03	3.1272E+03	**3.0115E+03**
	std	2.8303E+01	1.8368E+01	3.0958E+01	7.5187E+01	3.6630E+01	3.6654E+01	2.5238E+01	1.7528E+02	2.9560E+01	1.8278E+02	2.5361E+01	**1.3398E+01**
F26	ave	4.8545E+03	4.9846E+03	4.8538E+03	5.4630E+03	6.4673E+03	7.4331E+03	6.0053E+03	5.9947E+03	5.5235E+03	1.2477E+04	5.1153E+03	**4.3323E+03**
	std	4.6102E+02	1.6186E+03	4.5077E+02	1.2808E+03	5.2349E+02	1.6216E+03	6.0309E+02	4.6218E+02	**4.1459E+02**	2.0006E+03	2.4575E+03	6.5106E+02
F27	ave	3.3799E+03	3.3146E+03	3.3844E+03	3.5313E+03	3.4994E+03	3.6209E+03	3.4715E+03	3.5947E+03	3.3694E+03	4.1033E+03	3.4864E+03	**3.2442E+03**
	std	9.1358E+01	7.9157E+01	4.9943E+01	6.8005E+01	1.0577E+02	1.4868E+02	1.1132E+02	7.6837E+01	6.2087E+01	3.5665E+02	1.0420E+02	**1.2572E+01**
F28	ave	3.2918E+03	3.3504E+03	3.4484E+03	3.6442E+03	3.4165E+03	3.3469E+03	4.2288E+03	4.0835E+03	3.3360E+03	4.1598E+03	3.4168E+03	**3.2741E+03**
	std	2.4370E+01	3.7038E+01	4.5614E+01	1.0427E+02	4.9161E+01	3.3568E+01	1.9239E+03	3.3742E+02	3.6541E+01	2.6236E+02	2.9721E+01	**9.8792E+00**
F29	ave	3.9062E+03	3.8215E+03	**3.6628E+03**	4.1961E+03	4.6089E+03	4.4025E+03	4.5941E+03	4.4012E+03	4.1472E+03	7.5310E+03	3.9021E+03	3.6897E+03
	std	3.4461E+02	2.7449E+02	2.4091E+02	2.2547E+02	3.2714E+02	4.1397E+02	2.6975E+02	3.3980E+02	3.8837E+02	1.0744E+03	3.7928E+02	**8.1528E+01**
F30	ave	1.0505E+06	9.5097E+05	1.3015E+06	1.5243E+06	3.7199E+06	9.3085E+05	2.6952E+06	7.7521E+06	1.2643E+06	1.4645E+08	**8.8876E+05**	8.8989E+05
	std	2.7089E+05	1.5639E+05	3.0124E+05	4.2271E+05	1.1618E+06	1.7018E+05	8.6989E+05	3.5674E+06	4.2252E+05	5.9745E+07	1.2175E+05	**8.1491E+04**

As shown in Table 8, the experimental results of the CEC2022 10-dimensional benchmark function demonstrate that MESNS exhibits superior optimization performance and good stability on this test platform. In the 10-dimensional case, MESNS achieves lower average values (ave) and smaller standard deviations (std) for most unimodal, multimodal, and mixed functions, indicating that the algorithm not only obtains highly accurate optimal solutions but also possesses strong operational stability and result consistency. Particularly on several complex multimodal and combined functions, MESNS shows a more significant accuracy advantage and lower fluctuation amplitude compared to other comparative algorithms, demonstrating a good balance between global exploration and local exploitation capabilities in complex search spaces. Furthermore, the overall result distribution shows that MESNS performs at a leading or competitive level on most functions, demonstrating strong comprehensive optimization capabilities. In conclusion, the CEC2022 10-dimensional test results further validate the effectiveness and stability of MESNS on different test platforms and for different function types, enhancing the experimental support for the algorithm's versatility and robustness.

Figure 6 shows the convergence curves, which visually reflect the dynamic optimization process of different algorithms on the CEC2022 test functions. Overall, MESNS exhibits faster convergence speed and higher final convergence accuracy on most test functions. In the early stages of iteration, the MESNS curve shows a significant decrease, rapidly reducing the objective function value within a few iterations, demonstrating strong global search capability and high search efficiency. In the later stages, its convergence process maintains a steady downward trend without significant oscillations or prolonged stagnation, indicating that the algorithm possesses good local exploitation and refined search capabilities in approaching the optimal solution. In contrast, some of the compared algorithms, while showing a certain downward trend in the initial stages, subsequently slowed down their convergence speed, and even exhibited premature convergence on complex multi-peak or combined functions, making it difficult to further improve the accuracy of the solution. Furthermore, on different types of functions, the MESNS convergence curve is generally smoother and has smaller fluctuations, further verifying the stability and robustness of its search process. In summary, the convergence trend analysis in Figure 6 further illustrates the comprehensive advantages of MESNS in terms of search efficiency, convergence speed, and solution quality, which corroborates the aforementioned statistical results.

The performance distribution results shown in Figure 7 can further evaluate the overall performance of different algorithms on the CEC2022 test functions from a statistical perspective. It can be seen that the distribution of MESNS's results on most functions is generally in the better range, with a more concentrated distribution of its objective function values and a lower median, indicating that the algorithm can stably obtain high-quality solutions on different types of test functions. Compared with other comparative algorithms, MESNS's performance distribution range is relatively narrow, with less dispersion and a lower frequency of extreme values, demonstrating strong consistency and robustness. However, some algorithms exhibit a large distribution span and obvious long-tail phenomenon, indicating significant performance fluctuations on complex multimodal or combined functions. Furthermore, the overall distribution of MESNS shows relatively small changes between different function categories, indicating its good adaptability to changes in problem structure. In summary, the performance distribution characteristics shown in Figure 7 further verify the advantages of MESNS in solution quality, stability, and comprehensive optimization capabilities, consistent with the aforementioned analysis results of the mean, standard deviation, and convergence curve, fully demonstrating the effectiveness and reliability of this algorithm on the CEC2022 test platform.

Table 8. Performance Comparison of the Proposed Algorithms on the 10-Dimensional CEC2022 Benchmark.

ID	Metric	SAO	SBOA	ESC	GRO	CBSO	TLBO	ALA	SOO	PGA	SGA	SNS	MESNS
F1	ave	3.0000E+02	3.0000E+02	3.0613E+02	3.1704E+02	3.0000E+02	**3.0000E+02**	3.0000E+02	**3.0000E+02**	3.0000E+02	3.0000E+02	3.0000E+02	**3.0000E+02**
	std	2.3198E-09	1.4179E-07	1.1362E+01	4.0551E+01	3.7833E-07	5.5855E-14	7.5579E-10	**0.0000E+00**	2.6022E-05	6.9529E-04	1.0337E-02	**0.0000E+00**
F2	ave	2.3198E-09	1.4179E-07	1.1362E+01	4.0551E+01	3.7833E-07	5.5855E-14	7.5579E-10	**0.0000E+00**	2.6022E-05	6.9529E-04	1.0337E-02	**0.0000E+00**
	std	7.7733E-02	7.5938E-01	7.4712E-01	2.0498E+00	2.5373E-02	3.5662E-02	3.6759E-02	4.2492E-01	5.0459E-01	1.3555E+00	2.1612E+00	**1.4516E-12**
F3	ave	4.0032E+02	4.0106E+02	4.0347E+02	4.0256E+02	4.0001E+02	4.0003E+02	4.0001E+02	4.0050E+02	4.0403E+02	4.0457E+02	4.0263E+02	**4.0000E+02**
	std	1.8105E-05	1.1078E-05	8.8259E-06	7.9247E-03	1.7266E-01	7.3823E-03	7.6111E-05	2.8715E-04	9.2993E-06	2.9249E-02	1.3384E-03	**0.0000E+00**
F4	ave	7.7733E-02	7.5938E-01	7.4712E-01	2.0498E+00	2.5373E-02	3.5662E-02	3.6759E-02	4.2492E-01	5.0459E-01	1.3555E+00	2.1612E+00	**1.4516E-12**
	std	6.0319E+00	4.0766E+00	1.9461E+00	3.5477E+00	2.7073E+00	3.9194E+00	3.6662E+00	3.5465E+00	5.5012E+00	4.9827E+00	3.2645E+00	**5.4207E-01**
F5	ave	6.0000E+02	6.0000E+02	6.0000E+02	6.0001E+02	6.0019E+02	6.0000E+02	6.0000E+02	6.0000E+02	6.0000E+02	6.0006E+02	6.0000E+02	**6.0000E+02**
	std	6.4650E-01	1.6346E-02	5.7811E-07	1.1607E-01	1.7813E+00	9.8362E-01	8.2948E-02	2.0881E-01	1.6346E-02	2.0989E-01	1.1636E+00	**0.0000E+00**
F6	ave	1.8105E-05	1.1078E-05	8.8259E-06	7.9247E-03	1.7266E-01	7.3823E-03	7.6111E-05	2.8715E-04	9.2993E-06	2.9249E-02	1.3384E-03	**0.0000E+00**
	std	4.0951E+03	3.7961E+03	3.1962E+03	7.7504E+02	5.3271E-01	2.0239E+03	2.6001E+01	3.0372E+01	5.7088E+03	5.7563E+03	1.4644E+03	**6.2929E-02**
F7	ave	8.1121E+02	8.0889E+02	8.0431E+02	8.0886E+02	8.1116E+02	8.0897E+02	8.1186E+02	8.0614E+02	8.1154E+02	8.1399E+02	8.0904E+02	**8.0266E+02**
	std	9.6304E+00	7.6269E+00	8.8855E+00	6.1599E+00	7.0370E+00	8.5573E+00	5.2734E+00	9.3613E+00	1.0393E+01	7.3832E+00	8.2920E+00	**8.6536E-01**
F8	ave	6.0319E+00	4.0766E+00	1.9461E+00	3.5477E+00	2.7073E+00	3.9194E+00	3.6662E+00	3.5465E+00	5.5012E+00	4.9827E+00	3.2645E+00	**5.4207E-01**
	std	1.3656E+01	1.0292E+01	8.8647E+00	8.1191E+00	9.2044E+00	9.5992E+00	8.2018E+00	7.8661E+00	1.7253E+00	1.2818E+00	7.0691E+00	**4.2540E-01**
F9	ave	9.0012E+02	9.0000E+02	9.0000E+02	9.0004E+02	9.0063E+02	9.0028E+02	9.0002E+02	9.0008E+02	9.0000E+02	9.0012E+02	9.0064E+02	**9.0000E+02**
	std	1.4626E-13	3.8697E-13	1.7313E-02	2.5371E+01	2.0410E-04	4.6252E-13	1.8257E+01	1.8257E+01	2.9252E-13	8.1298E-03	8.1872E-13	**0.0000E+00**
F10	ave	6.4650E-01	1.6346E-02	5.7811E-07	1.1607E-01	1.7813E+00	9.8362E-01	8.2948E-02	2.0881E-01	1.6346E-02	2.0989E-01	1.1636E+00	**0.0000E+00**
	std	1.9390E+01	1.7287E+01	5.0620E-01	1.7287E+01	1.7287E+01	1.7287E+01	1.7287E+01	1.4895E+01	6.9999E+01	1.7285E+01	1.7287E+01	**5.3733E-12**
F11	ave	5.2228E+03	5.0150E+03	4.6468E+03	2.4036E+03	1.8013E+03	4.1126E+03	1.8202E+03	1.8351E+03	7.6031E+03	1.1093E+04	2.4986E+03	**1.8002E+03**
	std	2.3885E-13	4.2222E-13	1.7525E-04	9.4815E-08	2.7107E-04	4.8510E-13	5.4071E-13	**0.0000E+00**	3.4817E-13	6.7155E-03	1.1517E-12	**0.0000E+00**
F12	ave	4.0951E+03	3.7961E+03	3.1962E+03	7.7504E+02	5.3271E-01	2.0239E+03	2.6001E+01	3.0372E+01	5.7088E+03	5.7563E+03	1.4644E+03	**6.2929E-02**
	std	2.3864E-01	2.5226E-01	2.6018E-01	2.6669E-01	2.6784E-01	2.5039E-01	2.6742E-01	3.4257E-01	2.5572E-01	2.6343E-01	2.4667E-01	**1.4321E-02**

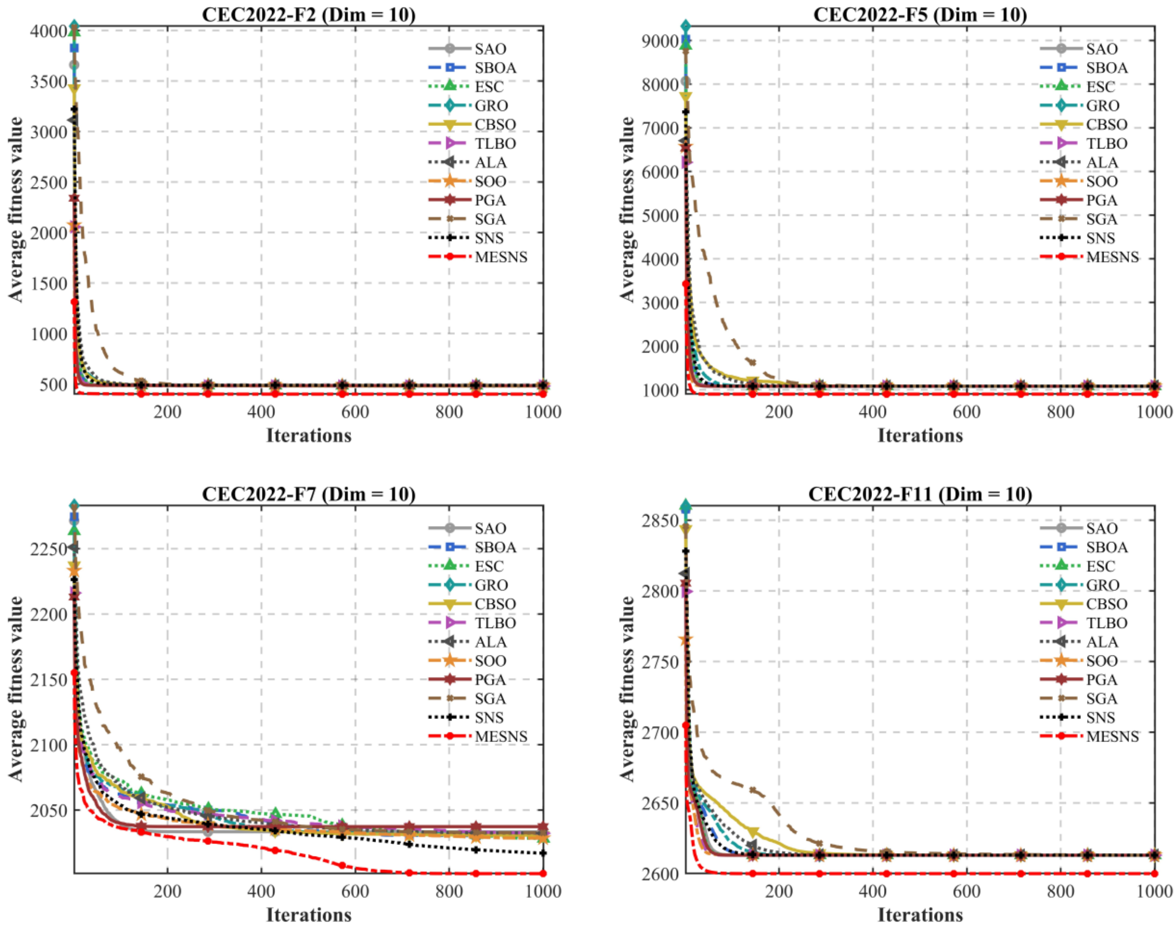
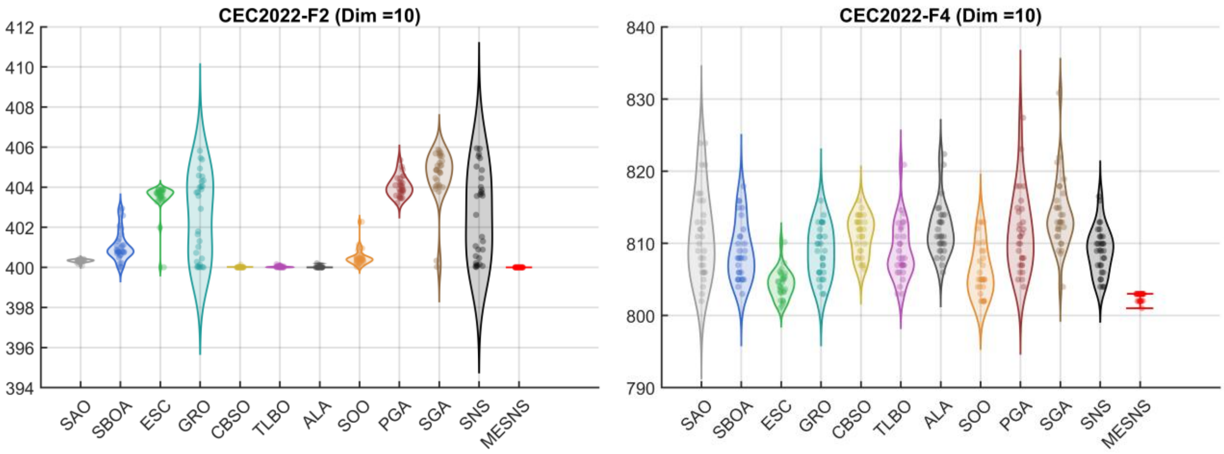


Figure 6. Convergence Speed Comparison of Different Algorithms on the CEC2022 Test Functions.



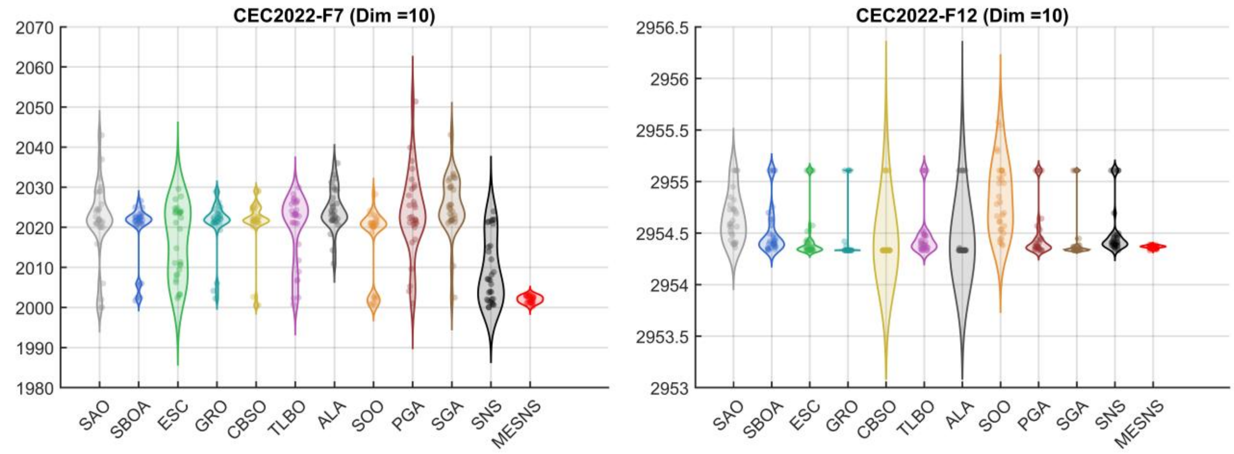


Figure 7. Performance Distribution of Different Algorithms on the CEC2022 Test Functions.

4.5. Statistical Analysis Based on the Friedman Test

To statistically examine the performance differences between the proposed MESNS algorithm and the competing methods [44], a Friedman mean rank analysis is conducted based on the experimental results obtained from the CEC2014, CEC2017, and CEC2022 benchmark test suites. The corresponding ranking outcomes are illustrated in Figure 8 and * indicates that the best ranking was achieved in each test set.

The overall ranking results shown in Figure 8 provide a unified evaluation of the performance of each compared algorithm across different benchmark test sets. It can be seen that MESNS ranks highest overall among all compared algorithms, with the lowest average ranking value, indicating that it maintains stable and excellent performance across multiple test platforms and different dimensional scales, including CEC2014, CEC2017, and CEC2022. This result demonstrates that MESNS not only excels in individual functions but also achieves competitive or even leading performance on most test functions, showcasing strong comprehensive optimization capabilities. In contrast, the rankings of other compared algorithms fluctuate relatively more, exhibiting some instability under different test sets or dimensional conditions, reflecting their higher sensitivity to changes in problem structure or dimensionality. Overall, the ranking analysis in Figure 8 further verifies the comprehensive advantages of MESNS in solution accuracy, convergence efficiency, and robustness from a global statistical perspective, providing more comprehensive and robust empirical support for the conclusion of algorithm superiority.

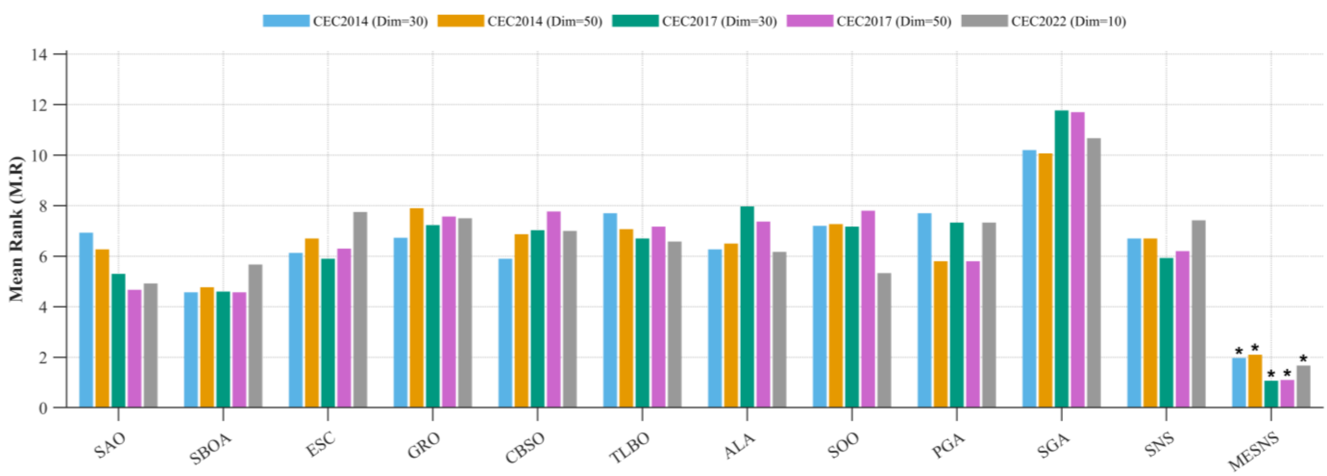


Figure 8. Ranking of each comparison algorithm.

4.6. Statistical Analysis Based on the Wilcoxon Signed-Rank Test

The performance of metaheuristic algorithms is inherently stochastic because their search processes rely on random initialization, probabilistic operators, and iterative exploration mechanisms. As a result, the optimization results obtained from different runs may vary significantly even under the same experimental settings. Therefore, evaluating algorithms only through average fitness values is insufficient to fully reflect their actual performance and stability. To provide a more reliable and statistically rigorous comparison, non-parametric statistical tests are widely adopted in the field of evolutionary computation and swarm intelligence. Among them, the Wilcoxon signed-rank test is one of the most commonly used methods because it does not require the data to follow a normal distribution and is suitable for pairwise comparisons between algorithms across multiple benchmark functions. By analyzing whether the performance differences between algorithms are statistically significant, the Wilcoxon test can effectively reduce the influence of randomness and provide stronger evidence for the superiority, robustness, and reliability of the proposed metaheuristic algorithm. Therefore, in this section, we conducted a Wilcoxon signed-rank test on the proposed algorithm, and the experimental results are shown below. Specifically, "W" represents the number of functions for which MESNS outperforms the comparison algorithm; "T" indicates instances where their performance is comparable; and "L" denotes the number of functions for which MESNS performs worse than its counterpart. As indicated by the W/T/L results presented in Table 9, MESNS demonstrates a significant advantage over the various comparative algorithms in the Wilcoxon signed-rank test. Specifically, on the CEC2014 test suite—even as the dimensionality increased to 50, resulting in a few "ties" or very rare "losses" on certain functions—MESNS maintained a clearly dominant position overall. For instance, compared to CBSO, it achieved results of 29/1/0 in both instances; compared to GRO, it attained 28/0/2 in both instances. This underscores the proposed algorithm's strong statistical superiority in addressing complex optimization problems of medium to high dimensionality. On the CEC2017 test suite, the advantages of MESNS were even more pronounced; under both 30-dimensional and 50-dimensional conditions, it achieved near-total victories against most comparative algorithms—with multiple comparisons yielding a perfect 30/0/0 record—and encountered only a negligible number of ties against algorithms such as GRO, SBOA, ESC, and SNS. This demonstrates that MESNS exhibits highly stable and significantly superior performance on this specific test platform. In the 10-dimensional tests within the CEC2022 suite, MESNS similarly maintained a strong competitive edge; for example, it achieved a perfect 12/0/0 record against both ESC and SGA, and attained 11/1/0 against both GRO and SNS. Although a small number of "losses" were recorded in comparisons against CBSO, SOO, and PGA, "wins" ("W") still constituted the overwhelming majority of the overall results. In summary, the results of the Wilcoxon signed-rank test conclusively demonstrate that MESNS consistently and significantly outperforms other algorithms across various test suites and dimensionality settings, thereby fully validating its effectiveness and superiority in terms of solution accuracy, stability, and overall robustness.

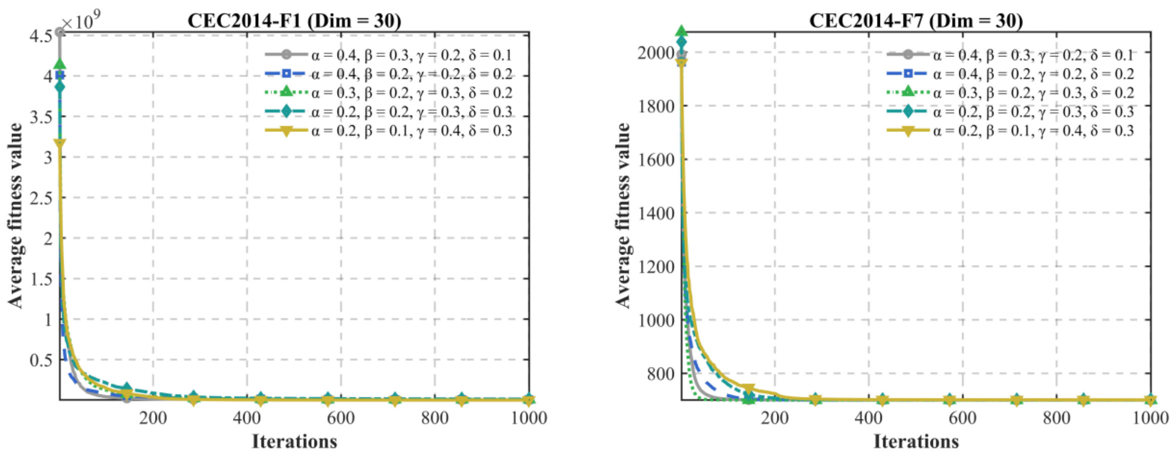
Table 9. W/T/L Results of the Compared Algorithms.

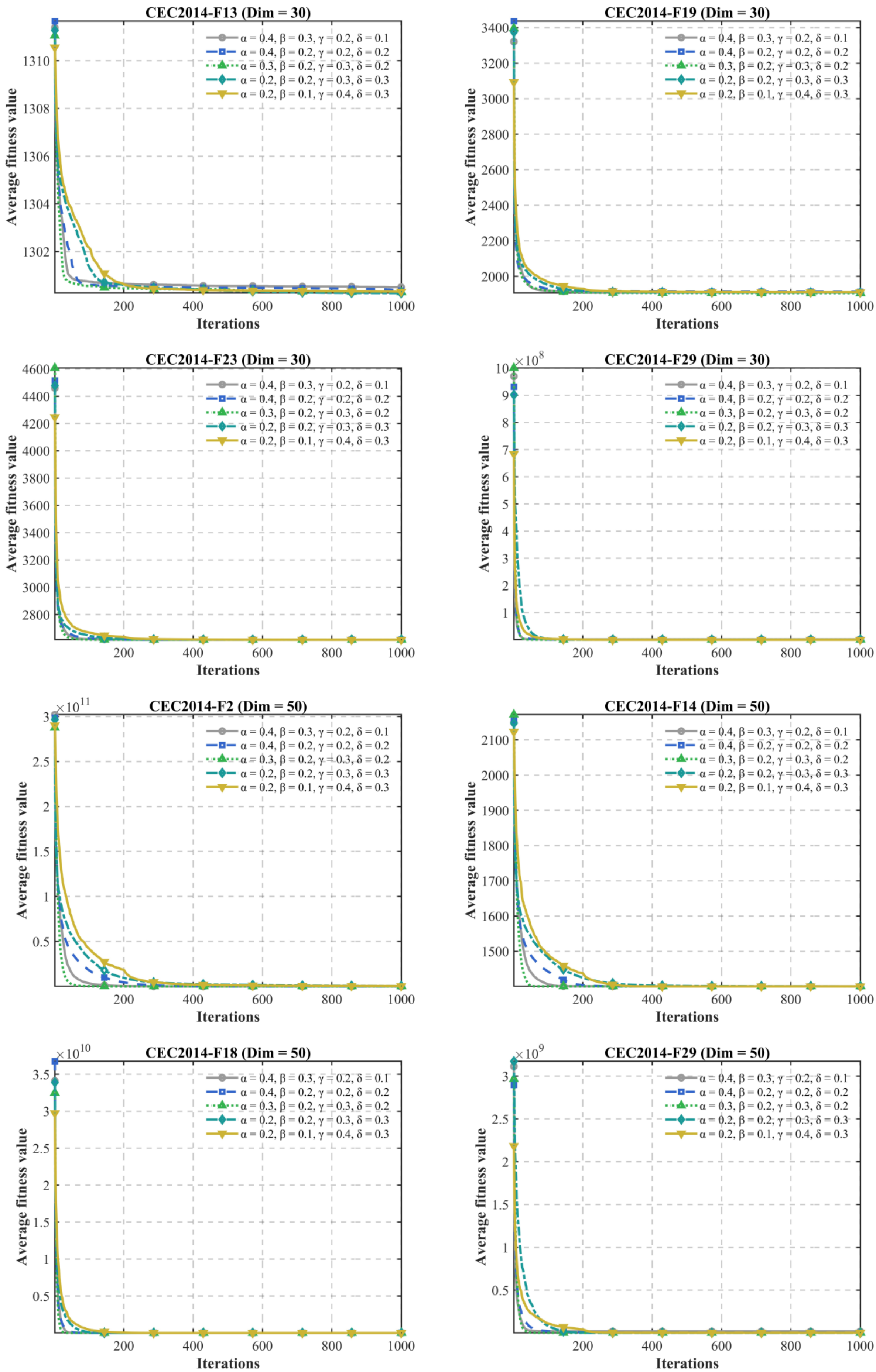
MESNS.VS	SAO	SBOA	ESC	GRO	CBSO	TLBO	ALA	SOO	PGA	SGA	SNS
CEC2014 (dim = 30)	28/2/0	26/0/4	26/1/3	28/0/2	29/1/0	28/0/2	25/1/4	27/0/3	30/0/0	25/2/3	27/0/3
CEC2014 (dim = 50)	27/3/0	26/0/4	26/0/4	28/0/2	29/1/0	28/0/2	25/0/5	26/1/3	28/1/1	25/1/4	26/1/3
CEC2017 (dim = 30)	30/0/0	30/0/0	30/0/0	29/1/0	30/0/0	30/0/0	30/0/0	30/0/0	30/0/0	30/0/0	29/1/0
CEC2017 (dim = 50)	30/0/0	29/1/0	28/2/0	30/0/0	30/0/0	30/0/0	30/0/0	30/0/0	30/0/0	30/0/0	29/1/0
CEC2022 (dim = 10)	8/4/0	9/3/0	12/0/0	11/1/0	11/0/1	8/4/0	10/2/0	8/3/1	8/2/2	12/0/0	11/1/0

4.7. Parameter Sensitivity Analysis

The performance of heuristic and metaheuristic algorithms is highly dependent on the selection of control parameters. Different parameter settings may significantly affect the balance between exploration and exploitation, thereby influencing the convergence speed, solution accuracy, and stability of the algorithm. Therefore, parameter sensitivity analysis is an essential step in evaluating the robustness and reliability of a proposed heuristic algorithm. By systematically varying the key parameters and observing their effects on optimization performance, it is possible to identify the most suitable parameter combinations and reveal how sensitive the algorithm is to different settings. Moreover, parameter sensitivity analysis can provide stronger evidence that the superiority of the proposed algorithm does not rely on a particular set of manually tuned parameters, but can be maintained across a reasonable range of values. In this section, we performed a parameter sensitivity analysis on the proposed MESNS, and the experimental results are shown below.

Figure 9 illustrates the sensitivity of MESNS to different parameter settings. It can be observed that the performance of the algorithm changes noticeably with the variation in parameter values, indicating that parameter selection has an important influence on the balance between global exploration and local exploitation. When the parameter values are set too small, the search behavior becomes relatively conservative, which may weaken the algorithm's ability to escape local optima and reduce the diversity of the population. In contrast, excessively large parameter values may lead to overly random search behavior, resulting in unstable convergence and a decline in solution accuracy. The experimental results show that MESNS achieves the best overall performance within an appropriate intermediate parameter range, where the algorithm can maintain both efficient exploration in the early stage and refined exploitation in the later stage. Moreover, the fluctuations of the results around the optimal parameter settings are relatively small, demonstrating that MESNS has good robustness and is not overly sensitive to slight parameter variations. Therefore, the selected parameter combination in this paper is reasonable and can effectively support the proposed algorithm in achieving stable and competitive optimization performance.




CEC2014-F13 (Dim = 30)
CEC2014-F19 (Dim = 30)
CEC2014-F23 (Dim = 30)
CEC2014-F29 (Dim = 30)
CEC2014-F2 (Dim = 50)
CEC2014-F14 (Dim = 50)
CEC2014-F18 (Dim = 50)
CEC2014-F29 (Dim = 50)
Average fitness value
Iterations
α = 0.4, β = 0.3, γ = 0.2, δ = 0.1
α = 0.4, β = 0.2, γ = 0.2, δ = 0.2
α = 0.3, β = 0.2, γ = 0.3, δ = 0.2
α = 0.2, β = 0.2, γ = 0.3, δ = 0.3
α = 0.2, β = 0.1, γ = 0.4, δ = 0.3

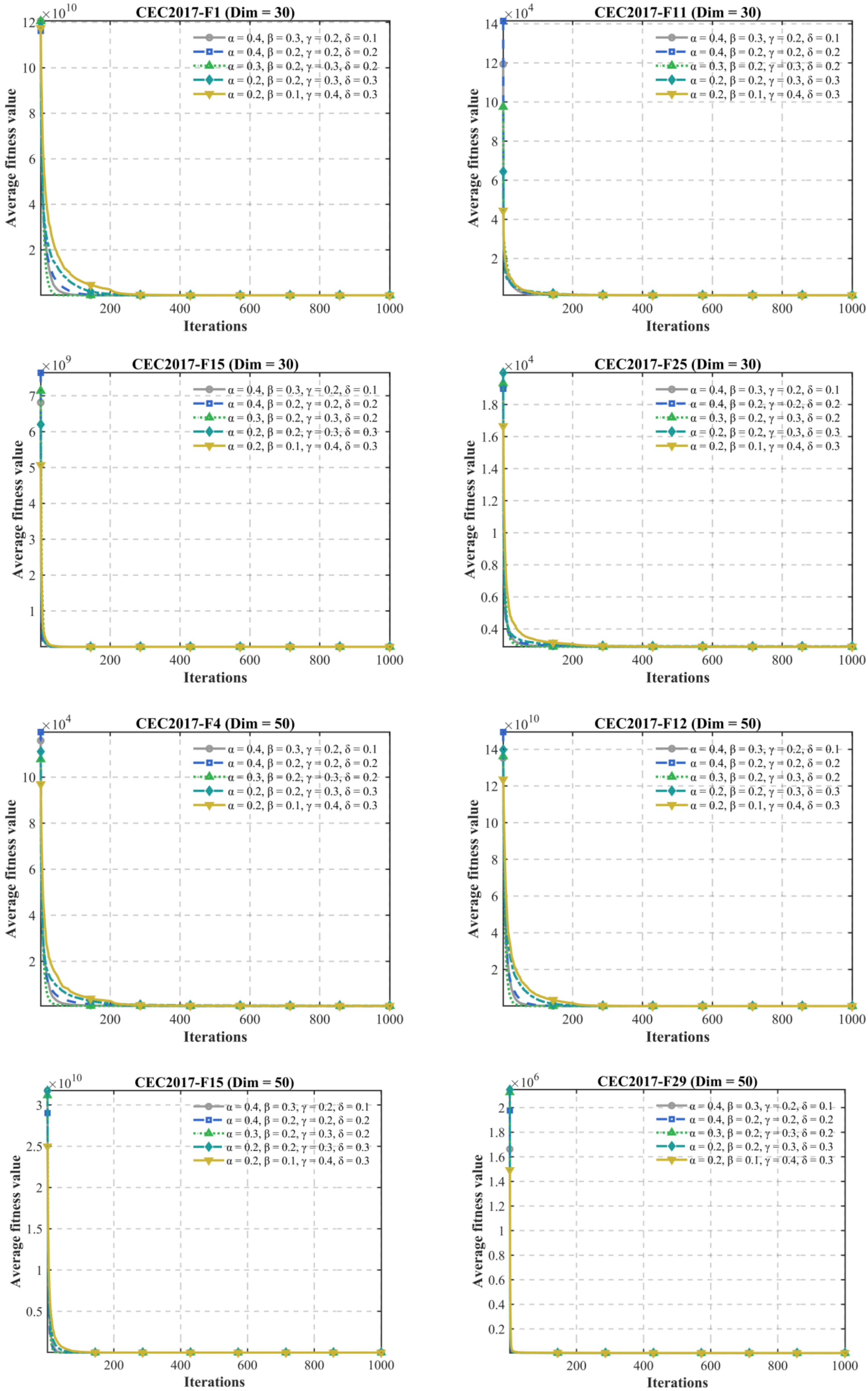

CEC2017-F1 (Dim = 30)
CEC2017-F11 (Dim = 30)
CEC2017-F15 (Dim = 30)
CEC2017-F25 (Dim = 30)
CEC2017-F4 (Dim = 50)
CEC2017-F12 (Dim = 50)
CEC2017-F15 (Dim = 50)
CEC2017-F29 (Dim = 50)
α = 0.4, β = 0.3, γ = 0.2, δ = 0.1
α = 0.4, β = 0.2, γ = 0.2, δ = 0.2
α = 0.3, β = 0.2, γ = 0.3, δ = 0.2
α = 0.2, β = 0.2, γ = 0.3, δ = 0.3
α = 0.2, β = 0.1, γ = 0.4, δ = 0.3
Average fitness value
Iterations

Figure 9. Parameter sensitivity analysis experimental results.

4.8. Population Diversity Analysis

Population diversity is an important indicator for evaluating the balance between exploration and exploitation in metaheuristic algorithms. A sufficiently high diversity level helps the population explore different regions of the search space and reduces the risk of premature convergence, while an excessively low diversity may cause the algorithm to become trapped in local optima, especially in high-dimensional and multimodal optimization problems. Since the proposed MESNS incorporates an elite-guided communication strategy, it is necessary to further investigate whether the enhanced exploitation capability significantly affects population diversity during the optimization process.

To quantitatively evaluate the population diversity, the average Euclidean distance between each individual and the population center is adopted as the diversity metric. Let the population at iteration t be denoted as $\{X_1(t), X_2(t), \dots, X_N(t)\}$, where N is the population size and D is the problem dimension. The population center can be defined as:

$$\bar{X}(t) = \frac{1}{N}\sum_{i=1}^{N} X_i(t) \tag{20}$$

Then, the population diversity at iteration t can be calculated as:

$$Div(t) = \frac{1}{N}\sum_{i=1}^{N} \|X_i(t) - \bar{X}(t)\| \tag{21}$$

where $\bar{X}(t)$ denotes the center of the population at iteration t, and $\|X_i(t) - \bar{X}(t)\|$ represents the Euclidean distance between the i-th individual and the population center. A larger value of $Div(t)$ indicates that the individuals are more dispersed in the search space, implying stronger exploration capability, whereas a smaller value indicates that the population is more concentrated, reflecting stronger exploitation behavior. To analyze the impact of the elite-guided communication strategy on population diversity, this paper compares MESNS against the original SNS and a variant without elite guidance, utilizing the CEC2017 test suite. Concurrently, population diversity curves spanning the entire optimization process were plotted and analyzed. The experimental results are presented below.

Figure 10 illustrates the population diversity curves of MESNS and SNS on several representative CEC2017 benchmark functions with 50 dimensions. Overall, SNS shows a very rapid decrease in diversity during the early iterations, and its diversity value quickly approaches zero on most functions. This indicates that the original SNS tends to lose population diversity prematurely and may become trapped in local optima, especially in high-dimensional problems. In contrast, MESNS maintains a relatively higher diversity level throughout the optimization process. Although its diversity also decreases gradually with the iteration progress, the reduction process is smoother and more stable.

For functions such as F4, F15, F19, and F28, MESNS preserves a significantly larger diversity than SNS in the middle and later stages of the search. This suggests that the proposed adaptive emotion selection mechanism, elite-guided communication strategy, and dynamic interaction radius adjustment effectively prevent the population from collapsing too early. In particular, on F15 and F19, the diversity of MESNS remains at a relatively high level for a long period, indicating that the algorithm still retains sufficient exploration capability even after many iterations. Meanwhile, SNS almost completely loses diversity after the early iterations, reflecting a stronger tendency toward premature convergence.

For functions such as F10 and F24, although the diversity of MESNS also decreases rapidly, its diversity remains higher than that of SNS before convergence. This indicates that MESNS can accelerate convergence while still maintaining a certain degree of exploration. Therefore, the results in Figure 10 demonstrate that MESNS achieves a better balance between exploration and exploitation than the original SNS. The proposed elite-guided communication mechanism enhances exploitation capability, while the retained random interaction mechanism helps preserve population diversity, thereby reducing the risk of premature convergence and improving the robustness of the search process.

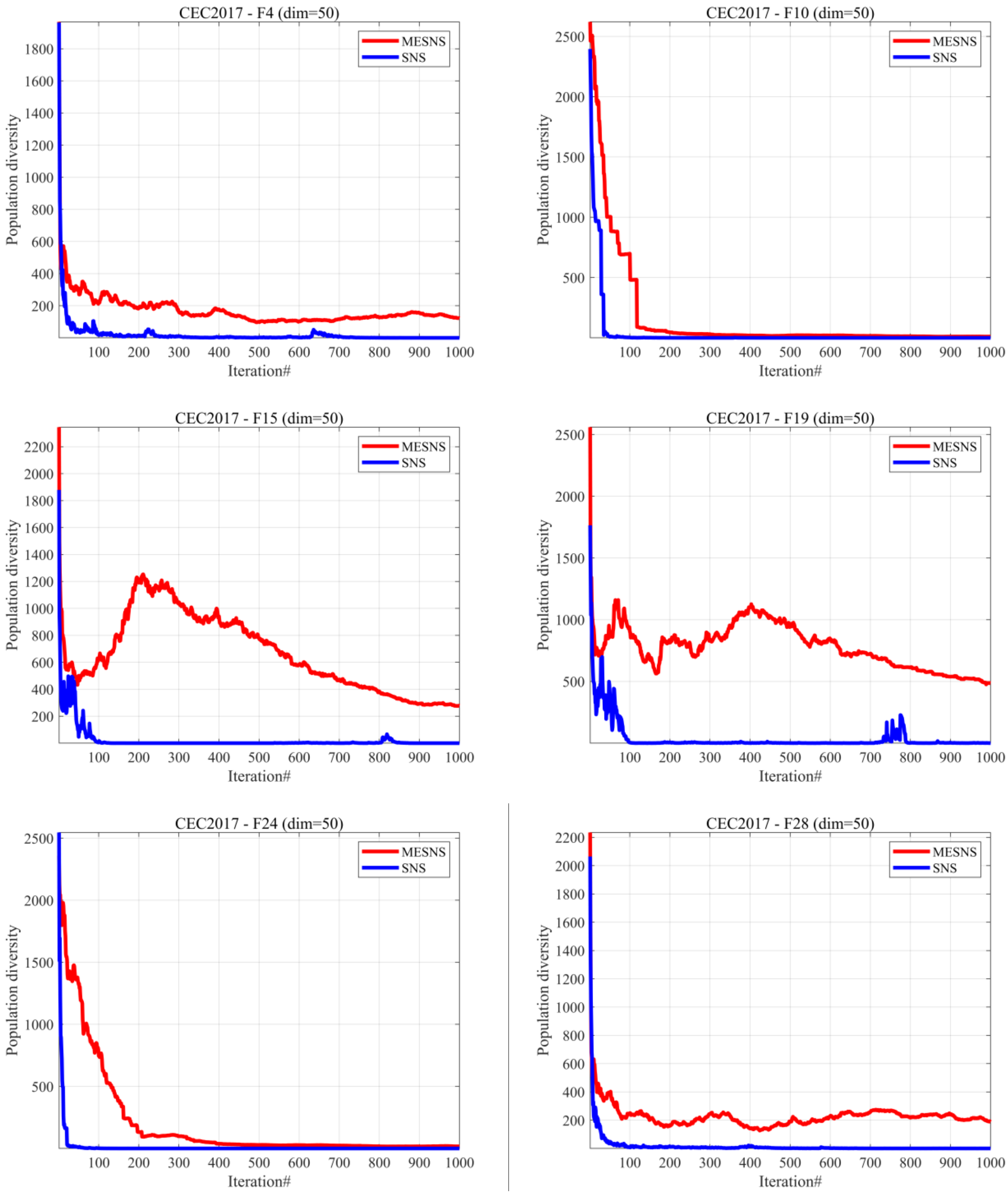


Figure 10. Experimental Results of Population Diversity Analysis.

5. Three-Dimensional Wireless Sensor Network Deployment Model

In practical wireless sensor network (WSN) applications, deployment optimization is closely related to localization performance, which is critical for tasks such as target tracking, environmental monitoring, and communication coordination. In particular, many localization problems can be formulated as nonlinear optimization problems, where the objective is to estimate node positions by minimizing measurement errors.

Recent studies have explored localization techniques based on Low Earth Orbit (LEO) satellite signals, where Doppler shift measurements are utilized to estimate receiver positions. In such approaches, the localization problem is typically modeled as a nonlinear optimization problem with respect to the unknown position variables. However, due to the lack of accurate initial solutions and the presence of satellite orbit errors and measurement noise, traditional methods often suffer from poor convergence or low accuracy.

To address these challenges, a particle swarm optimization-based Doppler shift correction (PSODC) technique has been proposed [45], where PSO is employed to obtain an initial coarse position by minimizing the Doppler frequency error, followed by orbit correction and fine positioning using iterative methods. This strategy significantly improves positioning accuracy and robustness in complex environments.

Inspired by these optimization-driven localization techniques, the WSN deployment problem considered in this paper can also be interpreted from a localization-oriented perspective. By improving the distribution of sensor nodes, the proposed MESNS algorithm not only enhances coverage but also has the potential to improve localization accuracy. Future work will further integrate localization models into the deployment optimization framework.

5.1. System Model and Assumptions

Wireless sensor networks deployed in complex environments, such as subsurface monitoring, underwater observation, and three-dimensional spatial surveillance, often require sensor nodes to operate within a volumetric region. In such scenarios, conventional two-dimensional deployment models are insufficient to accurately describe spatial coverage characteristics. The extension from two-dimensional to three-dimensional deployment leads to a substantial increase in the number of decision variables and results in intricate spatial coverage interactions, which significantly complicate the optimization process [46,47].

In this work, the monitored environment is modeled as a finite cubic region in a three-dimensional Cartesian coordinate system, expressed as

$$\mathbf{\Omega} = \{(x, y, z) \mid x, y, z \in [0, L]\} \tag{22}$$

where L represents the length of each side of the cubic region.

Assume that N identical sensor nodes are placed inside Ω. Each node is equipped with a uniform isotropic sensing device, and its sensing capability is characterized by a spherical sensing volume with radius R. The influence of communication constraints among sensor nodes is neglected, and only sensing coverage performance is considered in the deployment optimization.

The position of the i -th sensor node in three-dimensional space is denoted as

$$\mathbf{s}_i = [x_i, y_i, z_i], i = 1,2, \dots, N \tag{23}$$

Accordingly, the deployment of all sensor nodes can be jointly described by a 3N3N3N-dimensional vector

$$\mathbf{X} = (x_1, \dots, x_N, y_1, \dots, y_N, z_1, \dots, z_N), \tag{24}$$

which serves as the decision variable of the optimization problem.

To guarantee that the sensing region of each sensor node remains entirely within the monitored volume, the coordinates of all sensor nodes must satisfy the following constraints:

$$x_i, y_i, z_i \in [R, L-R], i = 1,2,\dots,N. \tag{25}$$

These constraints prevent sensing regions from extending beyond the boundary of the monitoring space.

5.2. Volumetric Coverage Measurement

For an arbitrary point $\mathbf{p} = (x, y, z)$ located within the monitoring region Ω, the Euclidean distance between this point and the i -th sensor node is defined as:

$$d_i(\mathbf{p}) = \parallel \mathbf{p} - \mathbf{s}_i \parallel_2 = \sqrt{(x-x_i)^2 + (y-y_i)^2 + (z-z_i)^2}. \tag{26}$$

A point $\mathbf{p}$ is regarded as being effectively sensed if at least one sensor node satisfies the condition

$$d_i(\mathbf{p}) \le R. \tag{27}$$

Since analytical evaluation of volumetric coverage in continuous three-dimensional space is computationally intractable, a discretization-based numerical approach is adopted. Specifically, the monitoring region Ω is uniformly partitioned into L^3 grid points. For each grid point, a binary coverage indicator function is defined as

$$M(x,y,z) = \begin{cases} 1,\text{if the point is within the sensing range of any node,} \\ 0,\text{otherwise.} \end{cases} \tag{28}$$

Based on this discretized representation, the volumetric coverage ratio corresponding to a given deployment $\mathbf{X}$ is calculated as

$$C(\mathbf{X}) = \frac{1}{L^3} \sum_{x=1}^{L} \sum_{y=1}^{L} \sum_{z=1}^{L} M(x,y,z). \tag{29}$$

The value of $C(\mathbf{X})$ reflects the proportion of the monitored volume that is effectively covered by the sensor network.

5.3. Optimization Model Formulation

The primary objective of the three-dimensional sensor deployment problem is to identify the spatial configuration of sensor nodes that maximizes the overall volumetric coverage of the monitoring region. This objective can be mathematically formulated as

$$\max_{\mathbf{X}} C(\mathbf{X}), \tag{30}$$

subject to the spatial constraints given in Equation (23).

Considering that most population-based and swarm intelligence optimization algorithms are inherently designed for minimization problems, the above objective is transformed into an equivalent minimization form:

$$\min_{\mathbf{X}} f(\mathbf{X}) = \frac{1}{C(\mathbf{X})}. \tag{31}$$

This reformulation preserves the original optimization goal while enabling seamless integration with commonly used metaheuristic optimization algorithms.

In the current deployment model, all sensor nodes are assumed to have the same sensing radius for simplicity and computational efficiency. This assumption is commonly adopted in many wireless sensor network deployment studies because it facilitates the formulation of the coverage model and enables fair performance comparison among optimization algorithms. However, in practical applications, different sensor nodes may exhibit heterogeneous sensing capabilities due to hardware differences, residual energy levels, environmental interference, or deployment altitude variations. In such cases, the sensing radius of the i-th sensor can be represented as R_i, where different nodes may have different coverage ranges. The corresponding coverage probability and deployment optimization model can then be extended to a variable-range sensing scenario. Although the present work focuses on the homogeneous sensing case, the proposed MESNS framework can be naturally extended to heterogeneous wireless sensor network deployment problems with variable sensing radii, which will be considered in future research.

5.4. Experimental Setup for 3D WSN Deployment Optimization

To assess the performance and robustness of the proposed MS-SBOA in solving wireless sensor network coverage optimization problems, a series of numerical simulation experiments are conducted in this section. The experiments aim to optimize the spatial placement of sensor nodes within a bounded three-dimensional monitoring region, with the objective of maximizing volumetric coverage.

To investigate the scalability of the proposed algorithm under different network sizes, two representative deployment scenarios are considered, involving networks composed of 30 and 50 sensor nodes, respectively.

Throughout all experiments, the monitoring environment is modeled as a finite three-dimensional space, and all sensor nodes are assumed to be homogeneous with identical sensing capabilities. The binary sensing model introduced in Sections 5.1–5.3 is employed to determine the coverage status of discretized monitoring points. To ensure a fair and reproducible comparison, all competing algorithms are implemented under the same experimental settings and parameter configurations.

Each algorithm is independently executed 30 times to alleviate the influence of stochastic factors, and the mean results are reported for performance evaluation. The termination condition for all experiments is defined by a fixed maximum number of iterations. The detailed parameter settings used in the experiments are summarized in Table 10.

Table 10. Parameter settings for three-dimensional WSN coverage optimization experiments.

Monitoring Area	Number of Nodes N	Sensing Radius R	Maximum Iterations T
50 m × 50 m × 50 m	30/50	10 m	1000

To further demonstrate the applicability of the proposed MS-SBOA to practical engineering optimization tasks, the three-dimensional WSN deployment problem is adopted as the application scenario. Deployment configurations with 30-node and 50-node networks are examined in detail. A comprehensive comparative study is performed against several benchmark algorithms from three key aspects: spatial rationality of node distribution, convergence behavior, and quantitative coverage performance. The experimental results are presented and analyzed in the following subsections.

5.4.1. Experimental Analysis for 30-Node Deployment

In this section, we set the deployment node to 30 for experimental analysis, and the experimental results are shown below.

As shown in Figure 11, the MESNS algorithm produces a more uniformly distributed three-dimensional deployment pattern compared with the other competing algorithms. The sensing spheres are evenly dispersed throughout the monitoring volume, including boundary and corner regions, effectively reducing large uncovered voids while avoiding excessive overlap among neighboring nodes. This balanced spatial configuration directly contributes to its superior volumetric coverage performance. As reported in Table 11, MESNS achieves the highest Mean coverage of 76.42%, the highest Median of 76.54%, and the lowest standard deviation (Std = 0.33), ranking first according to the Friedman test.

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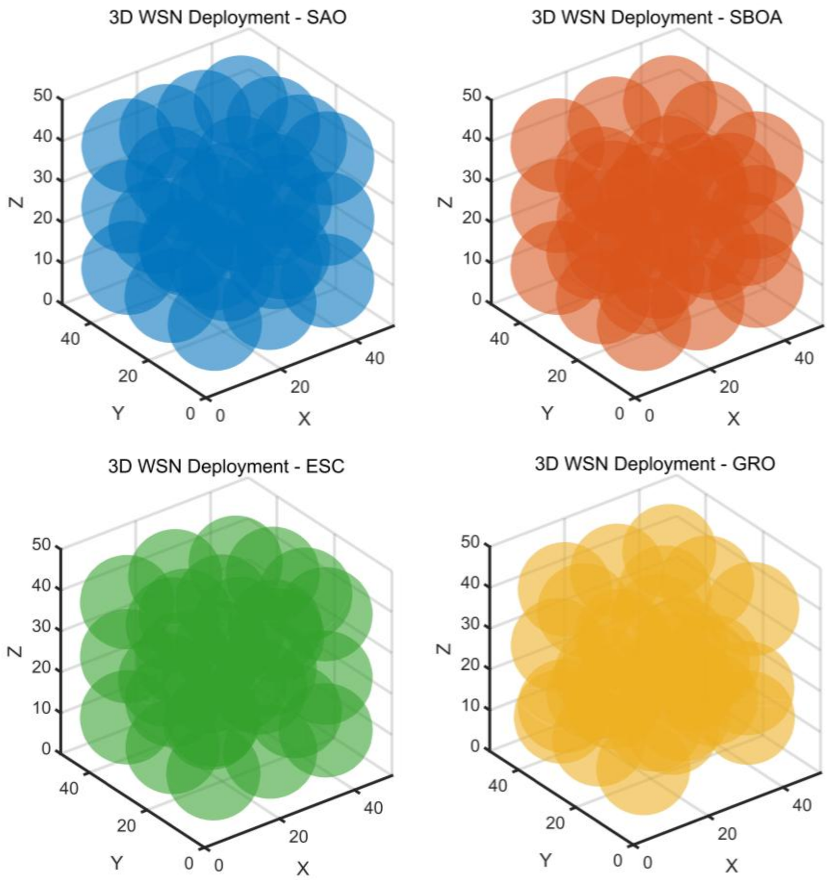
The small performance fluctuation further demonstrates its strong convergence stability and robustness. Compared with other algorithms that exhibit lower average coverage and higher variability, MESNS provides a more stable and efficient optimization solution for the 30-node three-dimensional WSN deployment problem.

As illustrated in Figure 12, the MESNS algorithm demonstrates superior convergence behavior in the 30-node three-dimensional WSN deployment scenario. MESNS exhibits a rapid increase in coverage during the early stages of iteration, indicating strong global exploration capability and high search efficiency. Compared with other algorithms, its convergence curve rises more steeply at the beginning and approaches a higher final coverage value with fewer iterations. In the later stages, the MESNS curve remains smooth and stable without significant oscillations or premature stagnation, reflecting its effective local exploitation and solution refinement capability. In contrast, several competing algorithms show slower improvement rates or early plateau phenomena, suggesting weaker ability to escape local optima. Overall, the convergence trend in Figure 12 confirms that MESNS achieves a better balance between exploration and exploitation, resulting in faster convergence speed, higher final coverage accuracy, and stronger optimization stability in the 30-node network deployment problem.



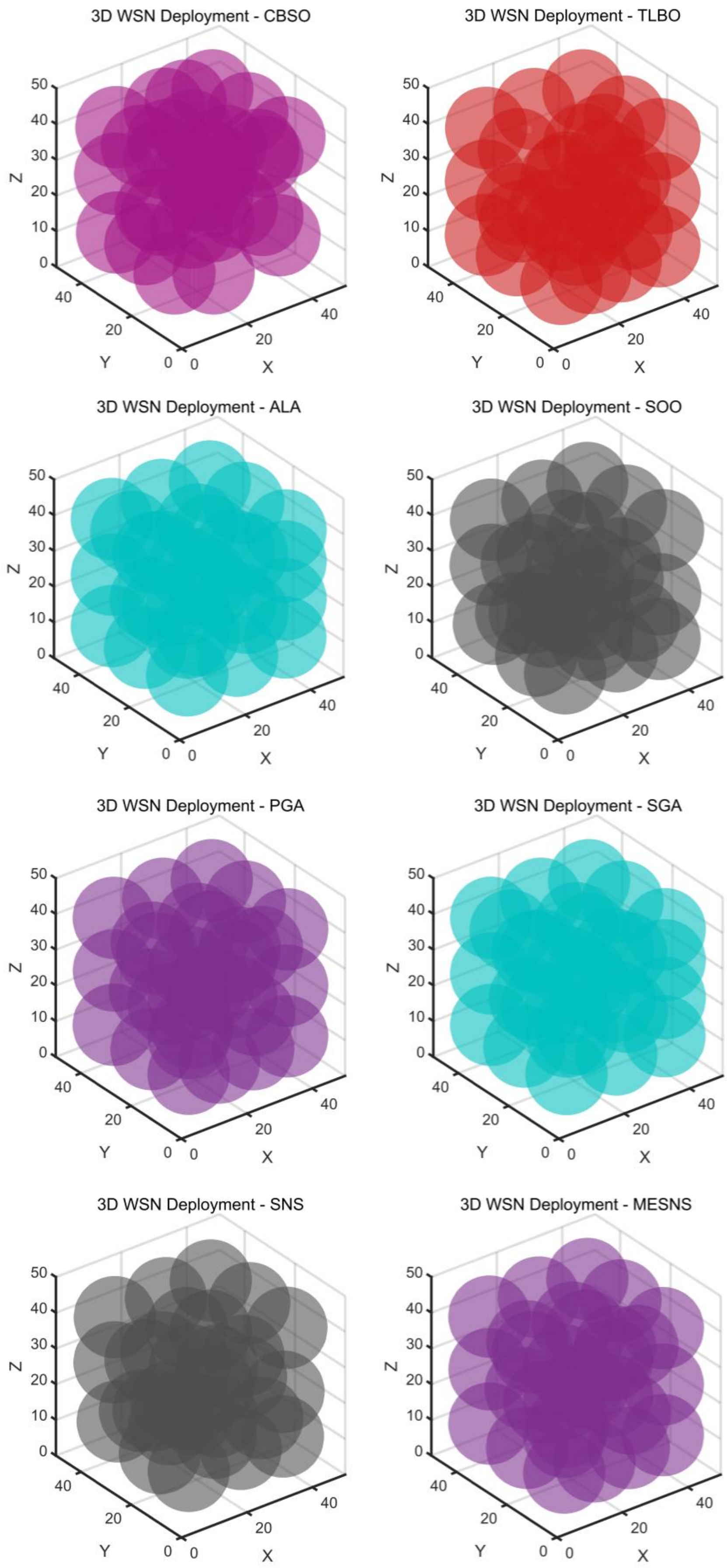


Figure 11. 3D node deployment distribution map obtained through different algorithms (30-Node).

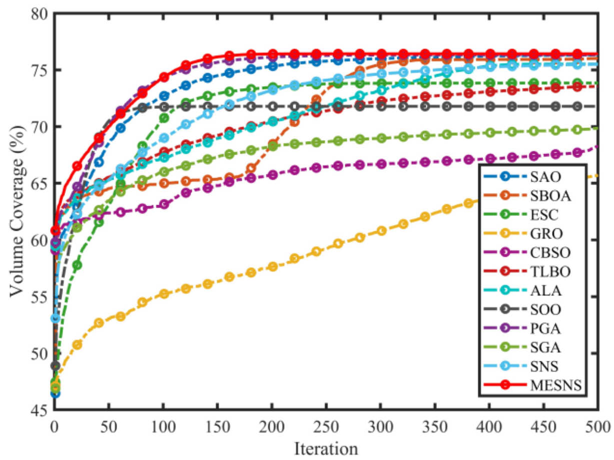


Figure 12. Convergence Performance of Different Algorithms in the 30-Node Network.

Table 11. Statistical Coverage Results of Different Algorithms on a 30-Node Network.

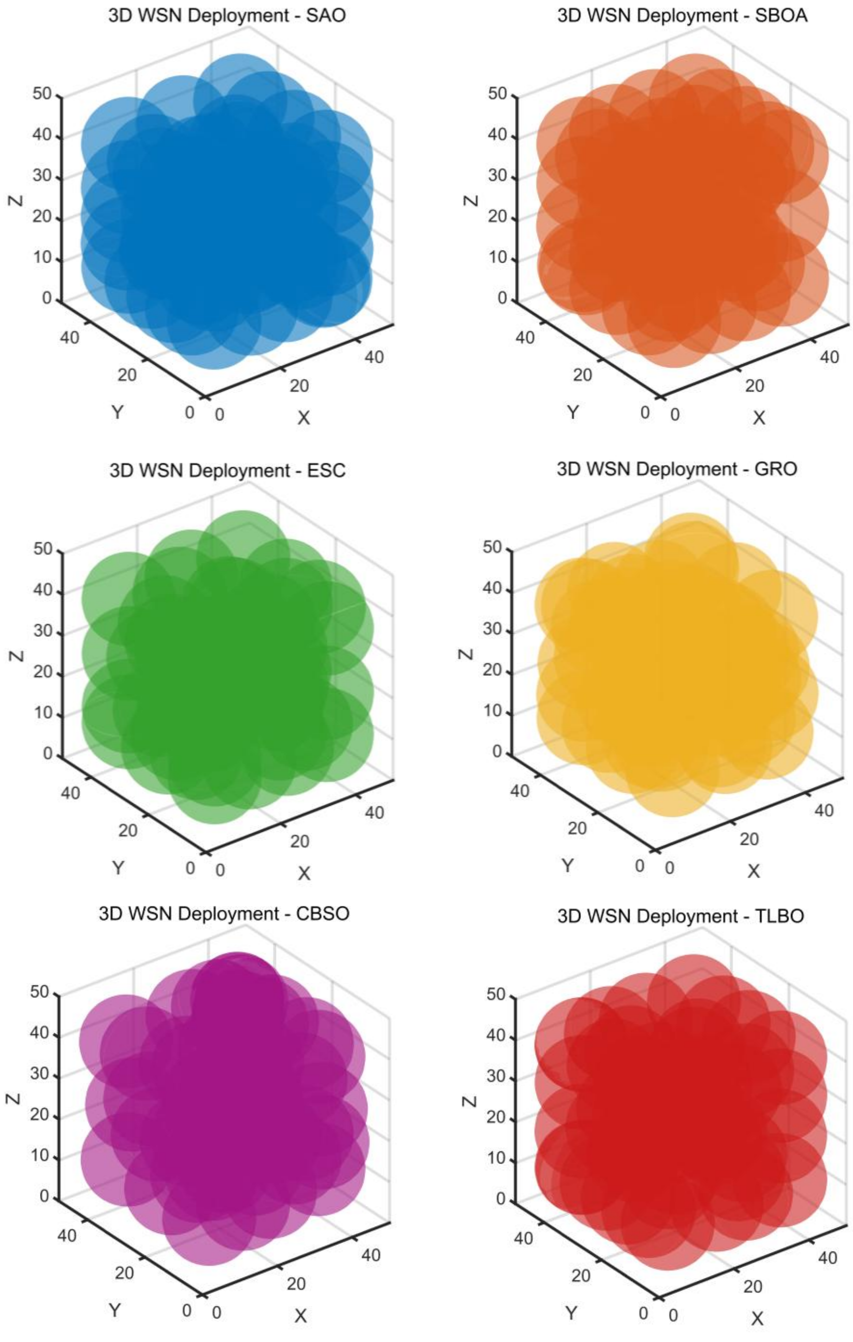
Algorithm	Best	Worst	Mean	Median	Std	Friedman Mean Rank
SAO	76.71	75.30	76.31	76.53	0.42	2.53
SBOA	76.52	74.23	76.01	76.25	0.57	4.00
ESC	75.32	72.24	73.84	73.91	0.79	7.40
GRO	68.01	62.34	65.71	66.06	1.37	11.93
CBSO	70.36	66.95	68.25	68.01	0.86	10.87
TLBO	75.90	70.01	73.55	73.84	1.83	7.40
ALA	76.53	73.72	75.52	75.47	0.77	4.90
SOO	73.15	69.54	71.78	71.78	0.82	8.93
PGA	76.71	75.18	76.32	76.46	0.42	2.63
SGA	72.77	65.27	69.85	70.12	1.72	10.03
SNS	76.12	74.06	75.49	75.58	0.43	5.27
MESNS	76.72	75.53	76.42	76.54	0.33	2.10

5.4.2. Experimental Analysis for 50-Node Deployment

In this section, we set the deployment node to 50 for experimental analysis, and the experimental results are shown below.

Figure 13 presents the three-dimensional node deployment distributions obtained by different algorithms in the 50-node WSN scenario. With the increase in network size, the spatial coverage interactions become more complex and the dimensionality of the decision space significantly increases, which places higher demands on the optimization algorithm. From the visual comparison, the deployment generated by MESNS exhibits a more uniform and well-balanced spatial arrangement across the entire monitoring volume, including boundary and corner regions. The sensing spheres are more evenly dispersed, effectively reducing redundant overlaps while minimizing uncovered gaps. In contrast, several competing algorithms display noticeable clustering or uneven density in certain subregions, leading to inefficient coverage allocation. This improved spatial rationality of MESNS is consistent with its superior statistical coverage performance in the 50-node case, where it achieves the highest Mean coverage (82.56%), competitive Best and Median values, and ranks first in the Friedman test. Overall, Figure 13 demonstrates that MESNS maintains strong scalability and spatial optimization capability as the network size increases, producing more efficient and stable deployment configurations in complex three-dimensional environments.

Figure 14 illustrates the convergence behavior of different algorithms in the 50-node three-dimensional WSN deployment scenario. As the network size increases, the optimization problem becomes more complex due to the higher dimensional decision space and intensified spatial coverage interactions, making convergence efficiency and stability more challenging to maintain. From the convergence curves, MESNS demonstrates a rapid improvement in coverage during the early iterations, reflecting strong global exploration capability and efficient search performance. Compared with other algorithms, it reaches a higher coverage level in fewer iterations and maintains a smooth and steadily increasing convergence trend throughout the optimization process. In the later stages, the MESNS curve shows no significant oscillation or premature stagnation, indicating robust local exploitation and refined solution adjustment. This behavior is consistent with the statistical results in Table 12, where MESNS achieves the highest Mean coverage (82.55%) and ranks first according to the Friedman test. Overall, Figure 14 confirms that MESNS preserves fast convergence speed, high final coverage accuracy, and strong optimization stability even in the more complex 50-node deployment scenario, demonstrating good scalability and robustness.



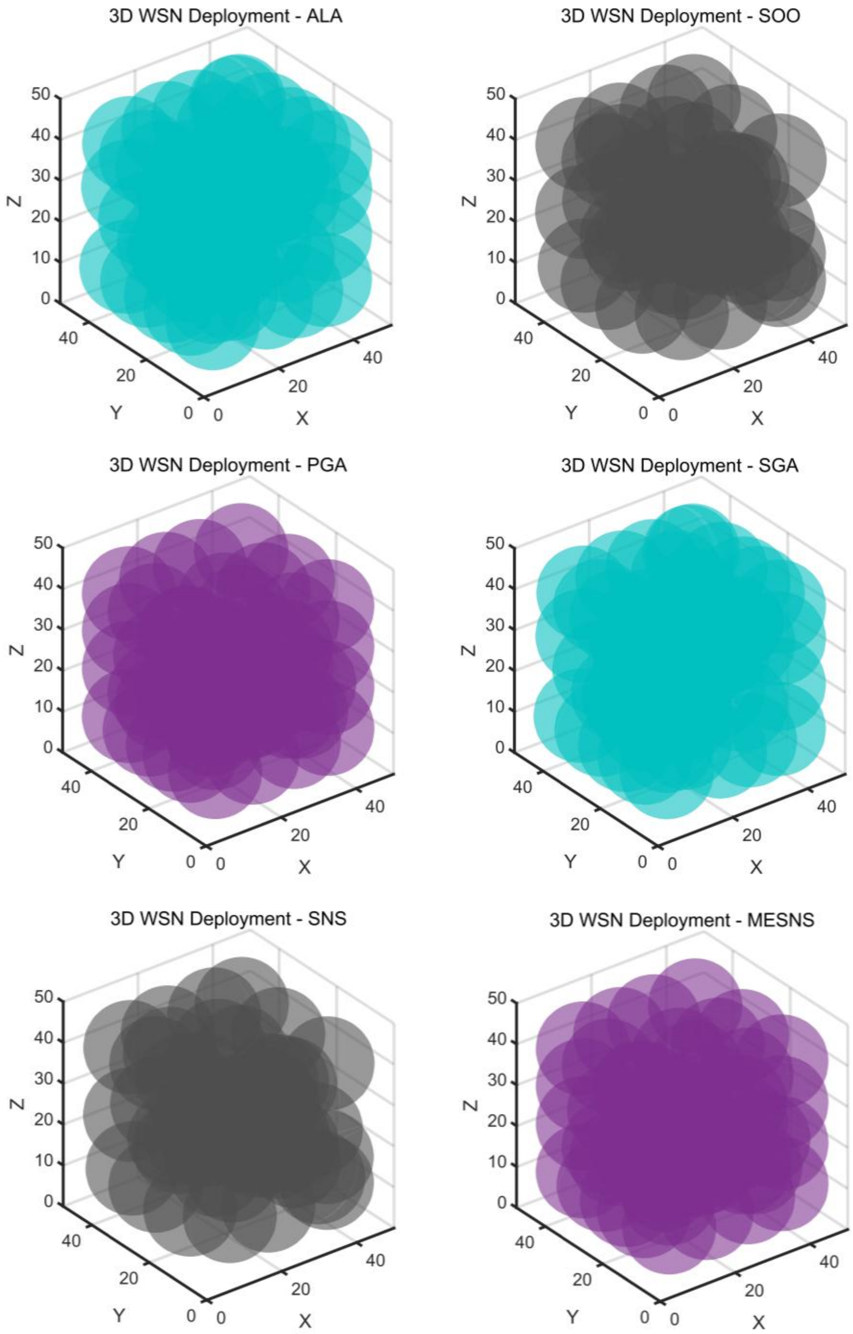


Figure 13. 3D node deployment distribution map obtained through different algorithms (50-Node).

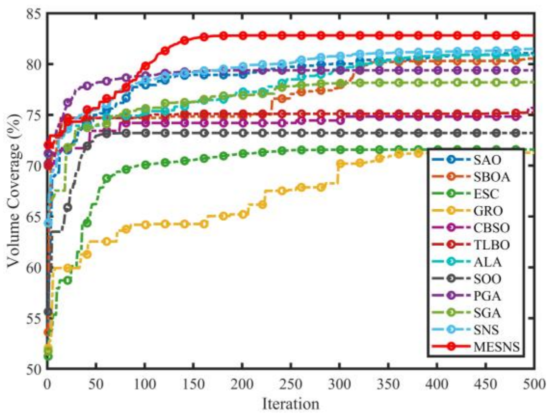


Figure 14. Convergence Performance of Different Algorithms in the 50-Node Network.

Table 12. Statistical Coverage Results of Different Algorithms on a 50-Node Network.

Algorithm	Best	Worst	Mean	Median	Std	Friedman Mean Rank
SAO	81.15	80.47	80.81	80.81	0.48	4.00
SBOA	81.89	81.03	81.46	81.46	0.61	3.00
ESC	72.35	72.27	72.31	72.31	0.06	10.50
GRO	69.98	69.25	69.61	69.61	0.52	12.00
CBSO	76.27	75.07	75.67	75.67	0.85	8.00
TLBO	79.39	76.94	78.17	78.17	1.73	6.50
ALA	82.16	81.50	81.83	81.83	0.47	2.50
SOO	74.50	72.25	73.37	73.37	1.59	10.50
PGA	80.30	79.38	79.84	79.84	0.65	6.00
SGA	75.38	74.55	74.96	74.96	0.58	9.00
SNS	80.61	80.00	80.30	80.30	0.43	5.00
MESNS	82.56	82.55	82.55	82.55	0.01	1.00

The optimized deployment strategy obtained by MESNS shows several important characteristics. First, the sensor nodes are distributed more uniformly throughout the monitoring space, which helps reduce coverage holes and improve the overall sensing efficiency. Second, excessive overlap among neighboring sensors is significantly reduced, allowing more effective utilization of limited sensing resources. Third, more sensor nodes tend to be allocated near sparsely covered or boundary regions, which further enhances the coverage quality of difficult-to-monitor areas.

In addition, the optimized deployment maintains a reasonable distance among sensor nodes, avoiding both excessive aggregation and overly sparse distribution. In the three-dimensional deployment scenario, the nodes are also distributed more appropriately across different height levels, which improves the spatial sensing capability of the network. These observations indicate that MESNS not only improves the numerical coverage rate but also generates a more balanced and practical deployment pattern for wireless sensor networks.

6. Conclusions and Future Work

In this paper, a multi-strategy enhanced social network search algorithm (MESNS) has been proposed to improve the search efficiency and optimization capability of the original SNS framework. By incorporating an adaptive decision-based emotion selection mechanism, an elite-guided communication strategy, and a dynamic interaction radius regulation mechanism, the proposed approach effectively enhances the balance between exploration and exploitation throughout the evolutionary process. The adaptive emotion selection mechanism enables stage-aware behavioral adjustment, while elite-guided information exchange accelerates convergence by promoting high-quality solution propagation. In addition, the dynamic interaction radius further strengthens global exploration in early iterations and refines local exploitation in later stages.

Extensive experiments conducted on IEEE CEC2014, CEC2017, and CEC2022 benchmark functions demonstrate that MESNS achieves superior performance in terms of optimization accuracy, convergence speed, and robustness compared with several state-of-the-art algorithms. Furthermore, its successful application to a three-dimensional wireless sensor network deployment problem verifies its practical effectiveness and engineering applicability. These results confirm that the proposed enhancements significantly improve the original SNS framework and provide a competitive metaheuristic for complex numerical optimization problems.

Although MESNS shows promising performance, several directions merit further investigation. First, adaptive parameter control mechanisms based on reinforcement learning or self-adaptive strategies could be explored to further enhance robustness across different problem landscapes. Second, extending MESNS to handle constrained, multi-objective, and large-scale optimization problems would broaden its applicability in real-world scenarios. Third, hybridization with other optimization paradigms, such as differential evolution or covariance matrix adaptation strategies, may further improve convergence speed and solution quality. Finally, applying MESNS to more complex engineering applications, such as energy management systems, intelligent transportation optimization, or deep neural network training, would provide deeper insights into its scalability and practical potential.

In the current study, the deployment optimization objective mainly focuses on maximizing the coverage rate of the wireless sensor network. However, in practical applications, energy consumption and deployment cost are also important factors that should be considered. For example, excessive node movement during deployment may increase energy consumption, while deploying a large number of sensor nodes may increase hardware and maintenance costs.

Therefore, the proposed deployment model can be further extended into a multi-objective optimization framework by jointly considering coverage quality, energy efficiency, communication cost, and deployment expense. In such a framework, the optimization objective can balance sensing performance and resource consumption, making the deployment strategy more practical for real-world applications. Although these factors are not explicitly considered in the current work, they will be investigated in future research.

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